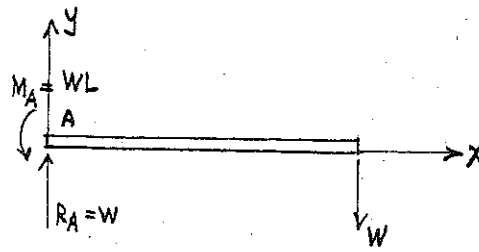
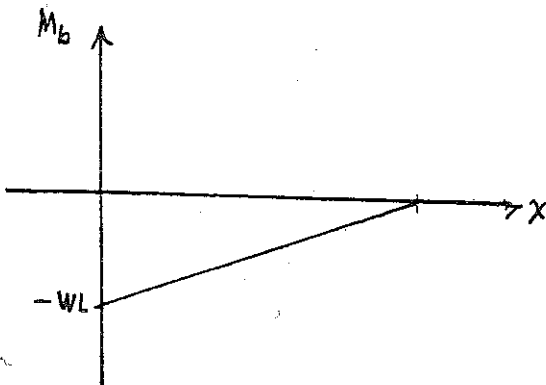
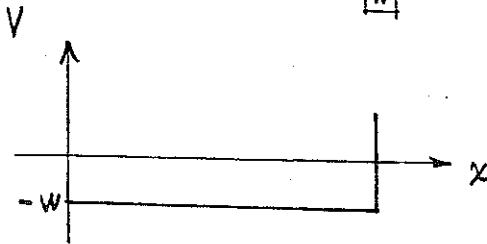
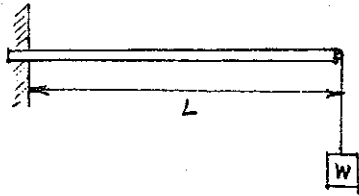


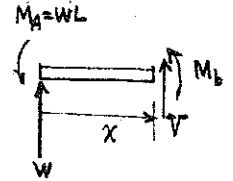
Chapter 3

(3-1)

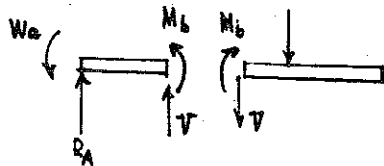
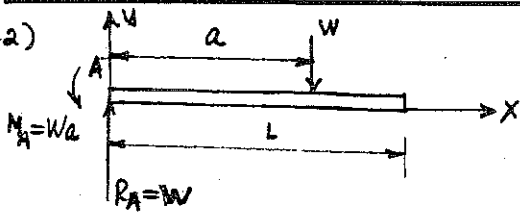


$$M_b = -M_A + Wx$$

$$= W(x - L)$$

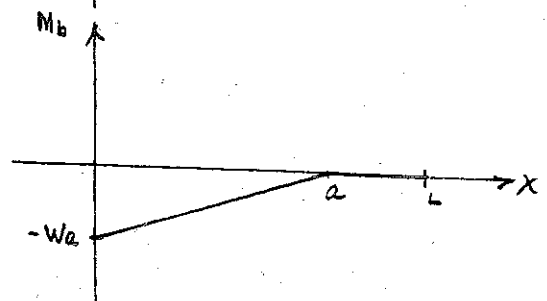
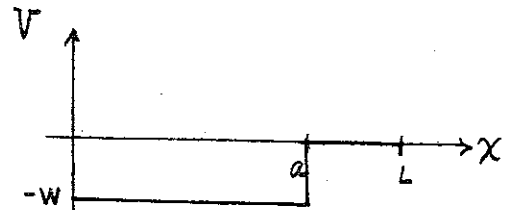


(3-2)

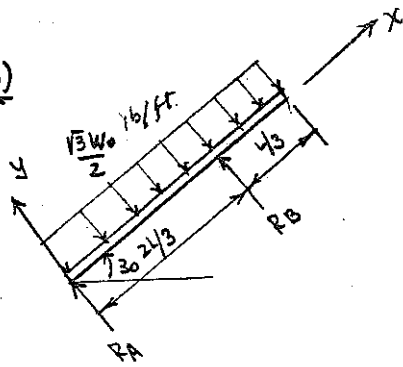


$$M_b = -W_A + R_A x - W(x - a)$$

$$= W(x - a) - W(x - a)$$



(3-3)



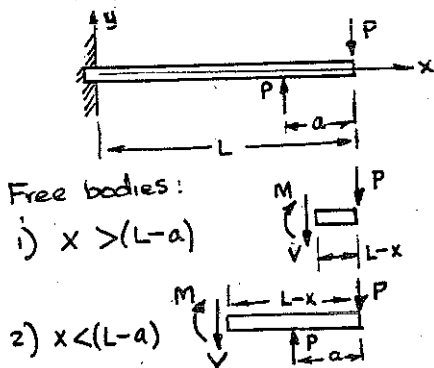
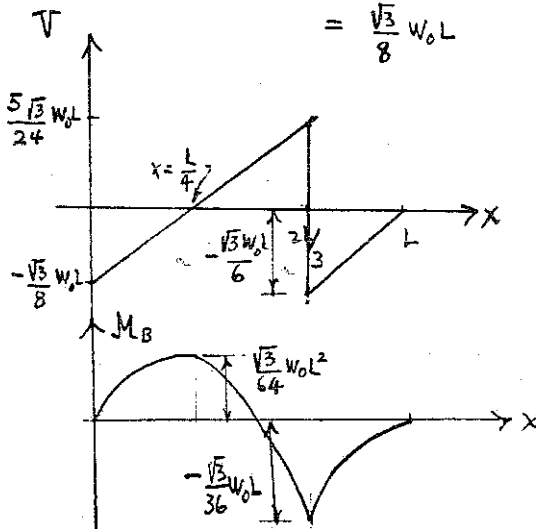
$$\sum F_y = 0$$

$$R_A + R_B = L W_0 \cos 30 = \frac{\sqrt{3}}{2} W_0 L$$

$$\sum M_A = 0 + \curvearrowright$$

$$R_B \frac{2L}{3} - \frac{\sqrt{3}}{2} W_0 \frac{L^2}{2} = 0$$

$$R_B = \frac{3\sqrt{3}}{8} L W_0 \rightarrow R_A = \frac{\sqrt{3}}{2} W_0 L - \frac{3\sqrt{3}}{8} W_0 L = \frac{\sqrt{3}}{8} W_0 L$$

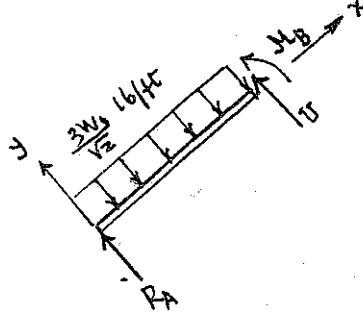


Free bodies:

1) $x > (L-a)$

2) $x < (L-a)$

$$(i) \quad 0 \leq x < \frac{2L}{3}$$



$$\sum F_y = 0$$

$$V(x) + R_A - \int_0^x \frac{\sqrt{3}}{2} W_0 dx = 0$$

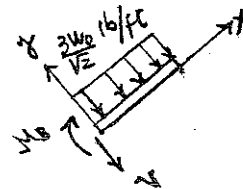
$$V(x) = -\frac{\sqrt{3}}{8} W_0 L + \frac{\sqrt{3}}{2} W_0 x = \frac{\sqrt{3}}{2} W_0 \left(x - \frac{L}{4} \right)$$

$$\sum M_z = 0 + \curvearrowright$$

$$M_B - R_A x + \int_0^x \frac{\sqrt{3} W_0}{2} x dx = 0$$

$$M_B = \frac{\sqrt{3} W_0 L}{8} x - \frac{\sqrt{3} W_0}{2} \frac{x^2}{2} = \frac{\sqrt{3}}{4} W_0 \left(\frac{Lx}{2} - x^2 \right)$$

(ii)



$$\frac{2L}{3} < x < L$$

$$\sum F_y = 0$$

$$V(x) + \int_0^{L-x} \frac{\sqrt{3} W_0}{2} dx = 0$$

$$V(x) = -\frac{\sqrt{3}}{2} W_0 (L-x)$$

$$\sum M = 0 + \curvearrowright$$

$$-M_B - \int_0^{L-x} \frac{\sqrt{3} W_0}{2} (L-x) dx = 0$$

$$M_B = -\frac{\sqrt{3}}{2} W_0 (L-x)^2$$

Convention: Positive M, V:

$$1) x > (L-a) : \sum F_y = 0 : V = -P$$

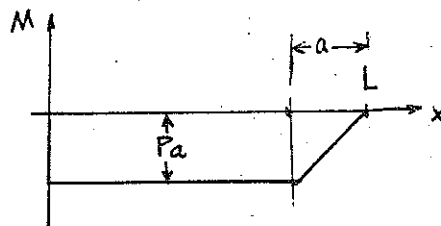
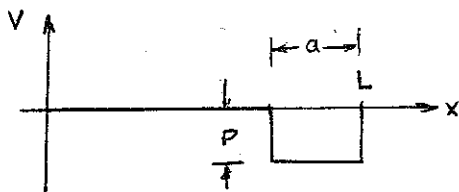
$$\sum M_z = 0 : -M - P(L-x) = 0 \therefore M = -P(L-x)$$

$$2) x < (L-a) : \sum F_y = 0 : -V + P - P = 0 \therefore V = 0$$

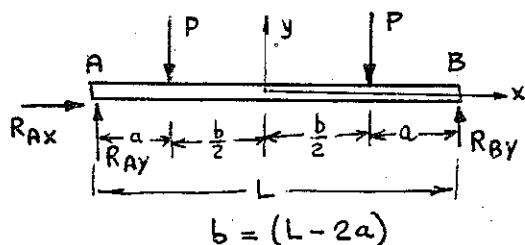
$$\sum M_z = 0 : -M - P(L-x) + P(L-x-a) = 0$$

$$\therefore M = -Pa$$

3.4 cont'd



3.5

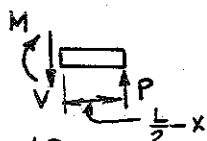


Sign Convention:

(positive)

Free body diagrams: ($x > 0$)

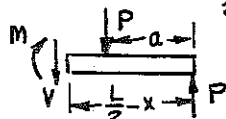
1) $x > \frac{b}{2}$



$$\sum F_y = 0 : \underline{V = P}$$

$$\sum (M_z) = 0 : \underline{M = P\left(\frac{L}{2} - x\right)}$$

2) $x < \frac{b}{2}$

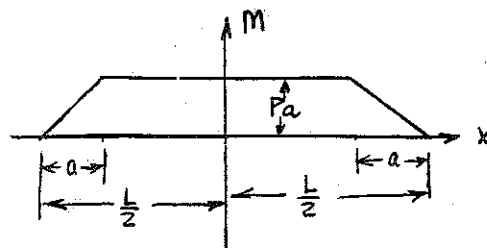
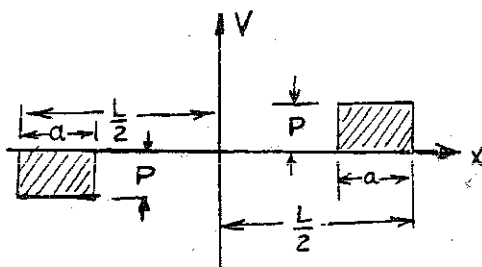


$$\sum F_y = 0 : -V - P + P = 0 \quad \underline{V = 0}$$

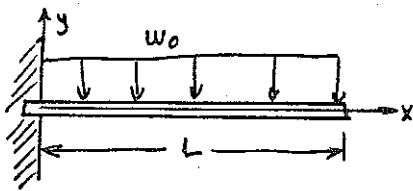
$$\sum (M_z) = 0 : -M - P\left(\frac{L}{2} - x - a\right) + P\left(\frac{L}{2} - x\right) = 0$$

$$\underline{M = Pa}$$

Similar treatment for $x < 0$



3.6

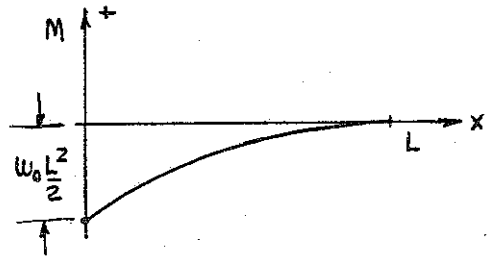
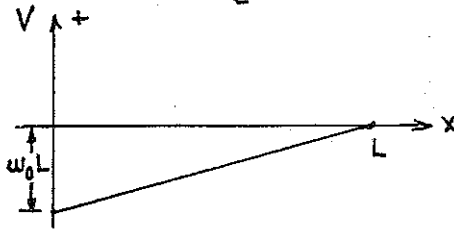


Positive V, M :

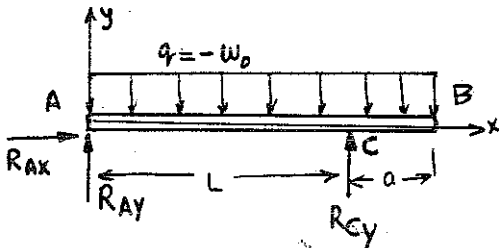
$$\frac{dV}{dx} = -w_0 \quad \therefore V = -w_0 x + C_1 \quad \text{At } x=L, V=0 \quad \therefore C_1 = w_0 L$$

$$\therefore V = w_0(L-x) \quad \therefore \frac{dM}{dx} = V = w_0(L-x)$$

$$\therefore M = \frac{w_0(L-x)^2}{2} + C_2 \quad M=0 \text{ at } x=L, \therefore C_2 = 0$$



3.7



No moment At A, no horizontal force at C.

Overall equilibrium:

$$\sum F_x = 0 : R_{Ax} = 0$$

$$\sum (M_z)_A = 0 : R_{Cy} L + (\text{moment of } q) = 0$$

$$\text{Moment due to } q \text{ is } \int_0^{L+a} (-w_0 dx)(x) = -\frac{w_0}{2}(L+a)^2$$

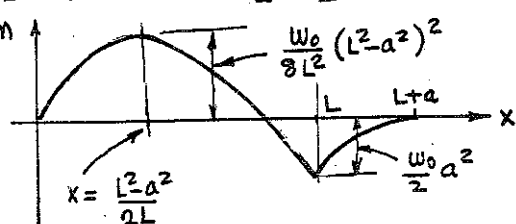
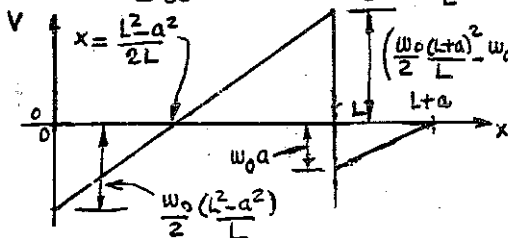
$$\therefore R_{Cy} = \frac{w_0}{2} \frac{(L+a)^2}{L}$$

$$\sum F_y = 0 : R_{Ay} + \frac{w_0}{2} \frac{(L+a)^2}{L} - w_0(L+a) = 0$$

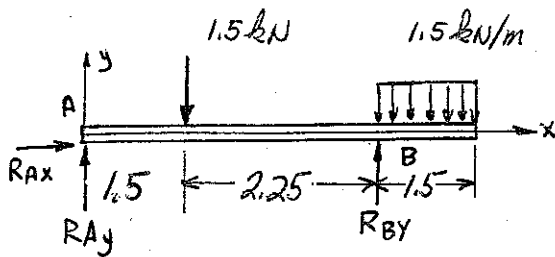
$$\therefore R_{Ay} = \frac{w_0}{2} \left(\frac{L^2 - a^2}{L} \right) \quad \text{Loading } q(x) = -w_0 \langle x \rangle^0 + \frac{w_0}{2} \frac{(L^2 - a^2)}{L} \langle x \rangle^1 + \frac{w_0}{2} \frac{(L+a)^2}{L} \langle x-L \rangle^1$$

$$V(x) = -\int_{-\infty}^x q(x) dx = \left\{ w_0 \langle x \rangle^1 - \frac{w_0}{2} \frac{(L^2 - a^2)}{L} \langle x \rangle^2 - \frac{w_0}{2} \frac{(L+a)^2}{L} \langle x-L \rangle^2 \right\}$$

$$M(x) = -\int_{-\infty}^x V(x) dx = \left\{ -\frac{w_0}{2} \langle x \rangle^2 + \frac{w_0}{2} \frac{(L^2 - a^2)}{L} \langle x \rangle^3 + \frac{w_0}{2} \frac{(L+a)^2}{L} \langle x-L \rangle^3 \right\}$$



3.8



No moment at A, No horizontal reaction at B.

$$\sum F_x = 0 : R_{Ax} = 0$$

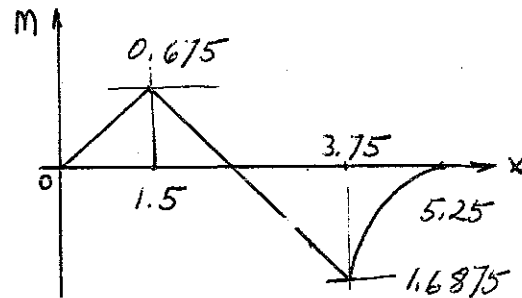
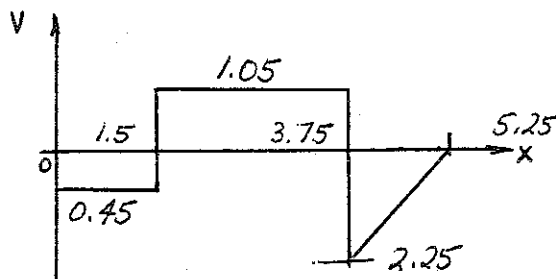
$$\sum (M_z)_A = 0 \Rightarrow 1.5(1.5) + R_{By}(3.75) - (1.5 \times 1.5)(4.5) = 0 \quad R_{By} = 3.3 \text{ kN}$$

$$\sum F_y = 0 \Rightarrow R_{Ay} = 0.45 \text{ kN}$$

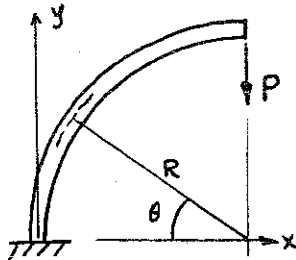
$$f(x) = \{ 0.45 \langle x \rangle_{-1} - 1.5 \langle x-1.5 \rangle_{-1} + 3.3 \langle x-3.75 \rangle_{-1} - 1.5 \langle x-3.75 \rangle^0 \}$$

$$V(x) = -0.45 \langle x \rangle_0 + 1.5 \langle x-1.5 \rangle^0 - 3.3 \langle x-3.75 \rangle^0 + 1.5 \langle x-3.75 \rangle^1 \text{ (kN)}$$

$$M(x) = 0.45 \langle x \rangle^2 - 1.5 \langle x-1.5 \rangle^2 + 3.3 \langle x-3.75 \rangle^2 - 0.75 \langle x-3.75 \rangle^2 \text{ (kN m)}$$



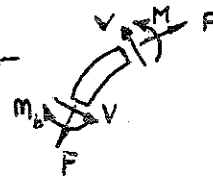
3.9



Convention for (M_b, F, V) :-

F = Normal (axial) force

V = Transverse (shear) force



$$\sum (M_z)_A = 0 : -M_b - P(R \cos \theta) = 0 \quad \underline{M_b = -PR \cos \theta}$$

$$\sum F_n = 0 \text{ (in axial direction at A) :}$$

$$F + P \cos \theta = 0$$

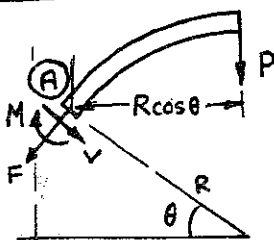
$$\underline{F = -P \cos \theta}$$

$$\sum F_t = 0 \text{ (in transverse direction at A) :}$$

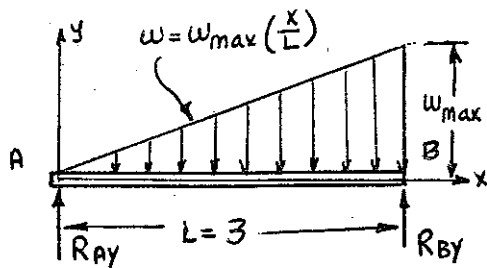
$$V + P \sin \theta = 0$$

$$\underline{V = -P \sin \theta}$$

Free body :



3.10



$$\text{Total load} = 24 \text{ kN} = \int_{x=0}^{x=L} w_{\max} \left(\frac{x}{L} \right) dx = w_{\max} \frac{L}{2}$$

$$\therefore w_{\max} = \frac{2}{L} 24 = 16 \text{ kN/m}$$

We wish to write $\sum (M_z)_A = 0$, and to do this we first find the

expression for the moment of the distributed load about A. The load on a differential element dx located at x is $w_{\max} \left(\frac{x}{L} \right) dx$. Its moment about A is $[-w_{\max} \left(\frac{x}{L} \right) dx (x)]$

$$\text{TOTAL MOMENT OF DISTRIBUTED LOAD} = -16 \int_0^3 \frac{x^2}{3} dx = -48 \text{ kN}\cdot\text{m}$$

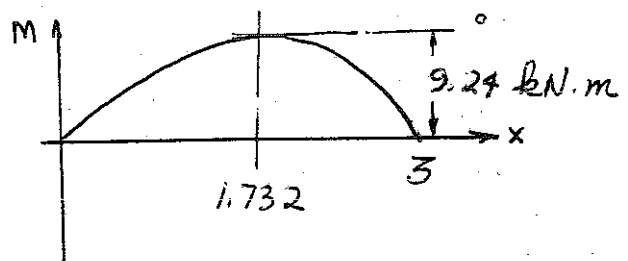
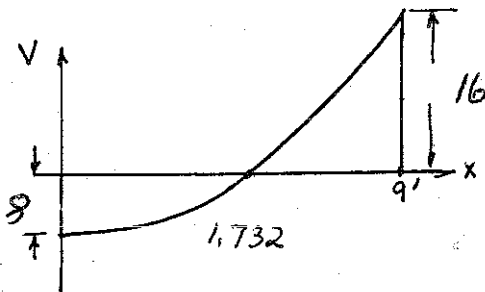
$$\sum (M_z)_A = 0 \quad -48 + 3 R_{By} = 0 \quad R_{By} = 16 \text{ kN}$$

$$\sum F_y = 0 \quad R_{Ay} + R_{By} - 24 = 0 \quad R_{Ay} = 8 \text{ kN}$$

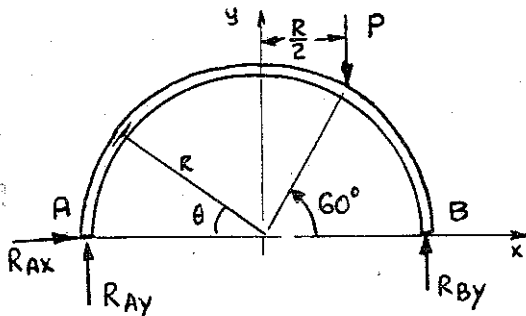
$$f(x) = 8 \langle x \rangle^{-1} - (w_{\max}/3) \langle x \rangle^1 + 16 \langle x-3 \rangle^{-1}$$

$$V = -8 \langle x \rangle^0 + (w_{\max}/6) \langle x \rangle^2 - 16 \langle x-3 \rangle^0$$

$$M = 8 \langle x \rangle^1 - (w_{\max}/18) \langle x \rangle^3 + 16 \langle x-3 \rangle^1$$



3.11



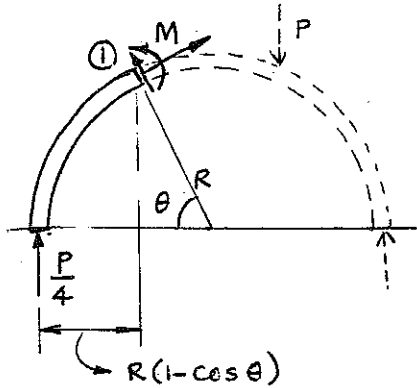
No moments at A, B ; no horizontal force at B.

$$\sum (M_z)_A = 0 : R_{By} (2R) - P \left(\frac{3R}{2} \right) = 0, \therefore R_{By} = \frac{3P}{4}$$

$$\sum F_x = 0 : R_{Ax} = 0$$

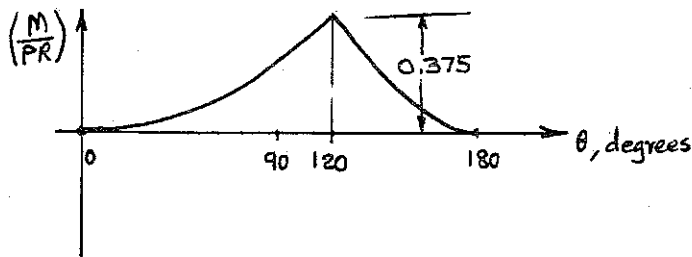
$$\sum F_y = 0 : R_{Ay} + R_{By} - P = 0 ; \therefore R_{Ay} = \frac{P}{4}$$

Free body, $\theta < 120^\circ$

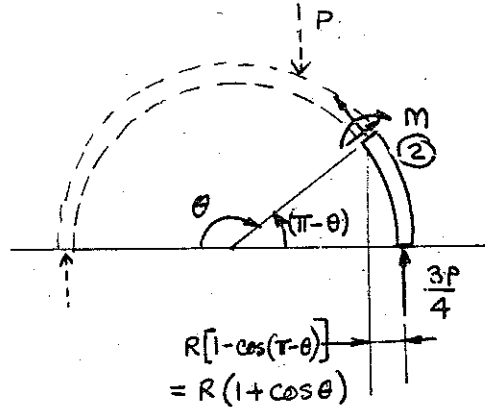


$$\sum (M_z)_1 = 0: M - \frac{P}{4}(R(1 - \cos \theta)) = 0$$

$$M = \frac{PR}{4}(1 - \cos \theta), \quad \theta < 120^\circ$$



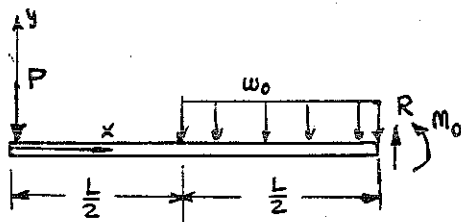
Free body, $\theta > 120^\circ$



$$\sum (M_z)_2 = 0: -M + \frac{3P}{4}R(1 + \cos \theta) = 0$$

$$M = \frac{3PR}{4}(1 + \cos \theta), \quad \theta > 120^\circ$$

3.12

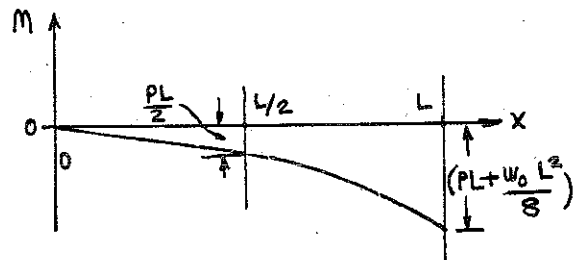
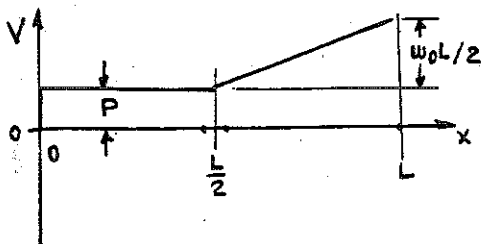


The reactions at the built in end will consist of a vertical force R and a moment M_0 . It is not necessary to find them.

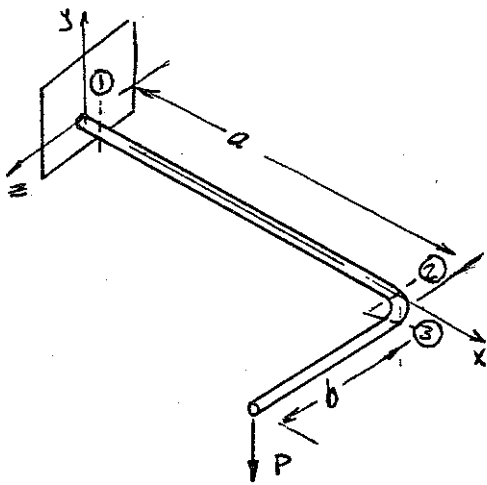
$$q(x) = -P\langle x \rangle_1 - w_0\langle x - \frac{L}{2} \rangle^0, \quad \text{good for } x < L$$

$$V = -\int_{-\infty}^x q dx = \left\{ P\langle x \rangle^0 + w_0\langle x - \frac{L}{2} \rangle^1 \right\} \quad (x < L)$$

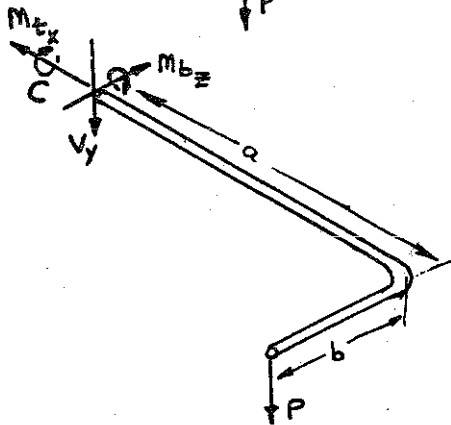
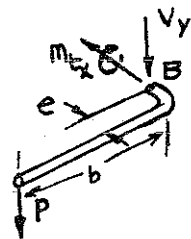
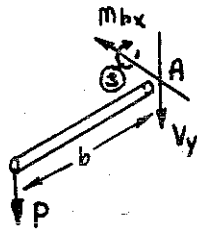
$$M = -\int_{-\infty}^x V dx = \left\{ -P\langle x \rangle^1 - \frac{w_0}{2}\langle x - \frac{L}{2} \rangle^2 \right\} \quad (x < L)$$



3.13



Free body diagrams :



$$\sum F_y = 0 : \underline{V_y = -P}$$

$$\sum (M_x)_A = 0 : -M_{bx} + P(b) = 0$$

$$\therefore \underline{M_{bx} = Pb}$$

Other components of $\vec{F}$ and $\vec{M}$ are zero.

$$\sum F_y = 0 : \underline{V_y = -P}$$

$$\therefore \sum (M_x)_B = 0 : \underline{M_{tx} = Pb}$$

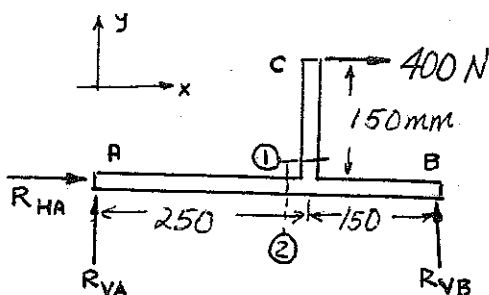
(Note : assume $e \ll b$ and $\therefore$ negligible).

$$\sum F_y = 0 : \underline{V_y = -P}$$

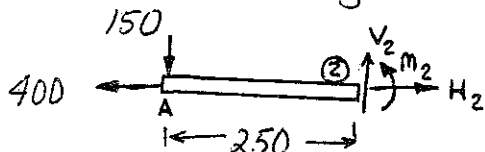
$$\sum (M_x)_C = 0 : \underline{M_{tx} = Pb}$$

$$\sum (M_z)_C = 0 : \underline{M_{bz} = -Pa}$$

3-14



Free body diagrams :-



Overall equilibrium: No moments at A and B, and no horizontal reaction at B due to the roller support.

$$\Sigma M_A = 0 : 400(150) - R_{VB}(400) = 0$$

$$\therefore R_{VB} = 150 \text{ N}$$

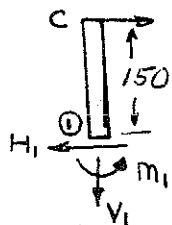
$$\Sigma F_y = 0 : R_{VA} + R_{VB} = 0 \therefore R_{VA} = -150 \text{ N}$$

$$\Sigma F_x = 0 : 400 + R_{HA} = 0 \therefore R_{HA} = -400 \text{ N}$$

$$\Sigma F_y = 0 : V_2 = 150 \text{ N}$$

$$\Sigma F_x = 0 : H_2 = 400 \text{ N}$$

$$\Sigma (M_z)_{\text{C}} = 0 : M_2 = -150(250 \times 10^{-3}) = 37.5 \text{ N}\cdot\text{m}$$

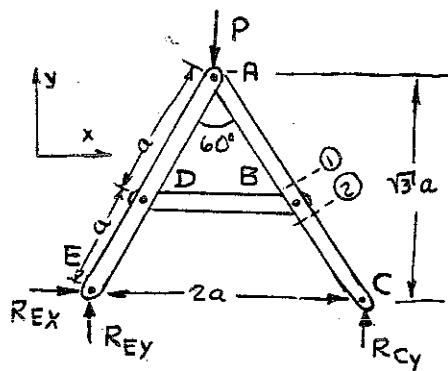


$$\Sigma F_y = 0 : V_1 = 0$$

$$\Sigma F_x = 0 : H_1 = 400 \text{ N}$$

$$\Sigma (M_z)_{\text{C}} = 0 : M_1 = 400(150 \times 10^{-3}) = 60 \text{ N}\cdot\text{m}$$

3-15



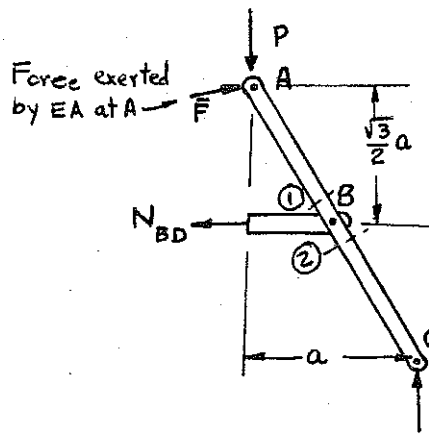
Forces and moments are in one plane (x-y). Assume: no moments at pin joints, no horizontal force at roller.

OVERALL EQUILIBRIUM: (NOTE: To solve the problem we only need the reaction at C and the forces in DB. We will write equations to find these forces only).

$$\Sigma (M_z)_E = 0 : R_{Cy}(2a) - P(a) = 0 \therefore R_{Cy} = \frac{P}{2}$$

The force in BD must lie along BD. To find it, we make a cut at the midpoint of BD and draw the following free body:-

3.15 CONT.

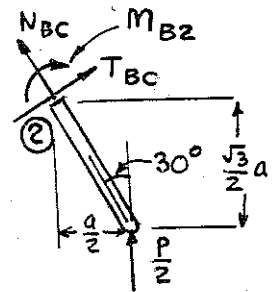


$$\sum (M_z)_A = 0 : \frac{P}{2}(a) - N_{BD}\left(\frac{\sqrt{3}}{2}a\right) = 0$$

$$\therefore N_{BD} = \frac{P}{\sqrt{3}}$$

Free body C② :

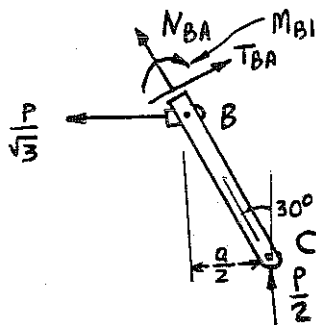
N: normal force
T: tangential force.



$$\sum (M_z)_{C②} = 0 : \frac{P}{2}\left(\frac{a}{2}\right) - M_{B2} = 0 \quad \therefore M_{B2} = \frac{Pa}{4}$$

$$\sum F = 0 \text{ along rod} : N_{BC} + \frac{P}{2} \cos 30^\circ = 0 \quad \therefore N_{BC} = -\frac{P\sqrt{3}}{4}$$

$$\sum F = 0 \text{ normal to rod} : T_{BC} + \frac{P}{2} \sin 30^\circ = 0 \quad T_{BC} = -\frac{P}{4}$$



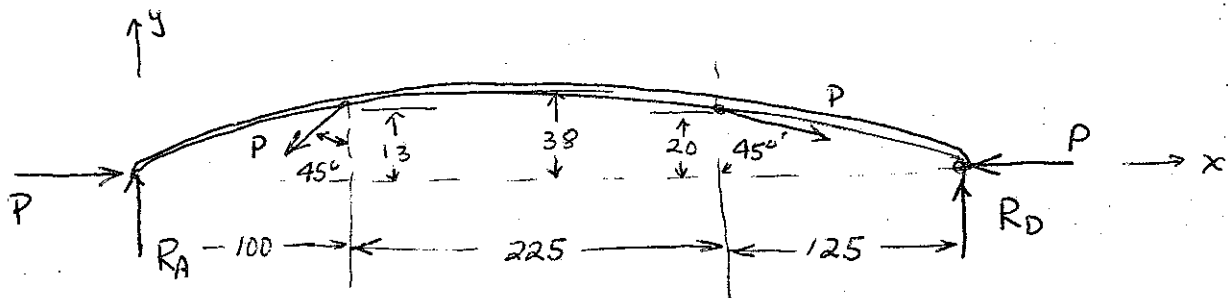
$$\sum (M_z)_{B①} = 0 : \frac{P}{2}\left(\frac{a}{2}\right) - M_{B1} = 0 \quad M_{B1} = \frac{Pa}{4}$$

$$\sum F = 0 \text{ along BC} : N_{BA} + \frac{P}{2} \cos 30^\circ + \frac{P}{\sqrt{3}} \cos 60^\circ = 0$$

$$\therefore N_{BA} = -\frac{\sqrt{3}}{4}P - \frac{P}{2\sqrt{3}} = -0.721P$$

$$\sum F = 0, \perp \text{ to BC} : T_{BA} + \frac{P}{2} \sin 30^\circ - \frac{P}{\sqrt{3}} \cos 30^\circ = 0$$

$$\therefore T_{BA} = -\frac{P}{4} + \frac{P}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} = \frac{P}{4}$$



DETERMINATION OF REACTIONS $\sum F_y = 0$ $R_A + R_D = \frac{2P}{\sqrt{2}}$

$$\sum M_A = 0 \Rightarrow \frac{P\sqrt{2}}{2} (13 - 100) + \frac{P\sqrt{2}}{2} (-20 - 325) + 450 R_D = 0$$

TO OBTAIN $R_D = 0.678 P$
 $R_A = 0.736 P$

SHEAR FORCES
OBTAINED BY
SECTIONS

$$0 < x < 100 \text{ mm}$$

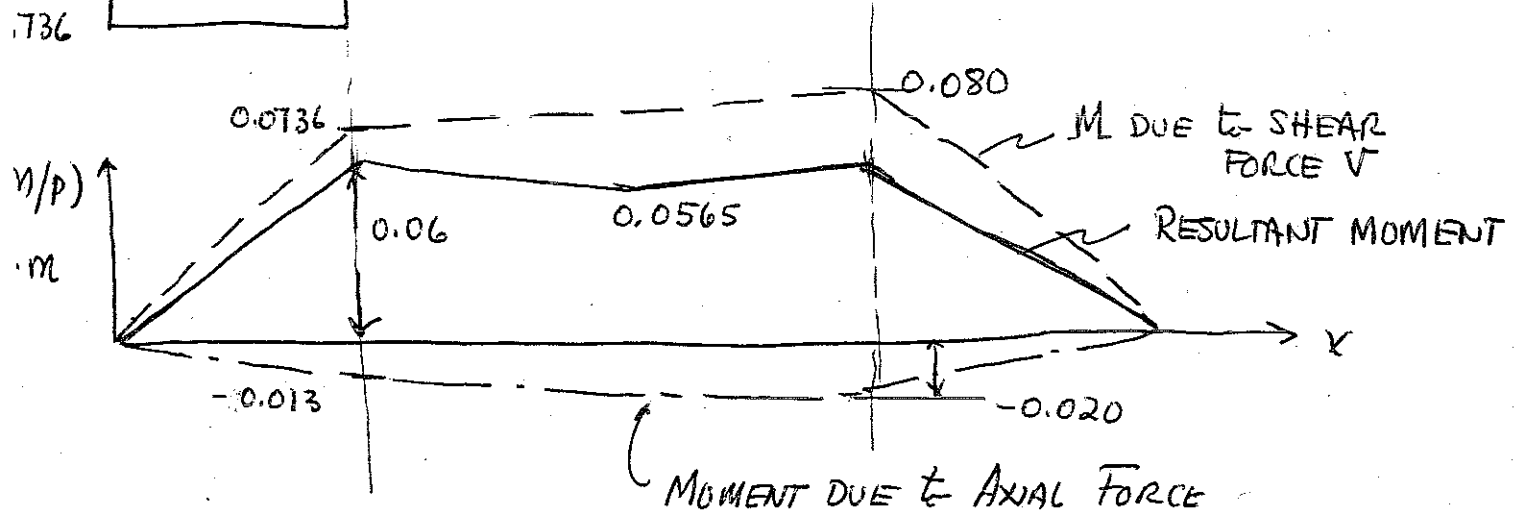
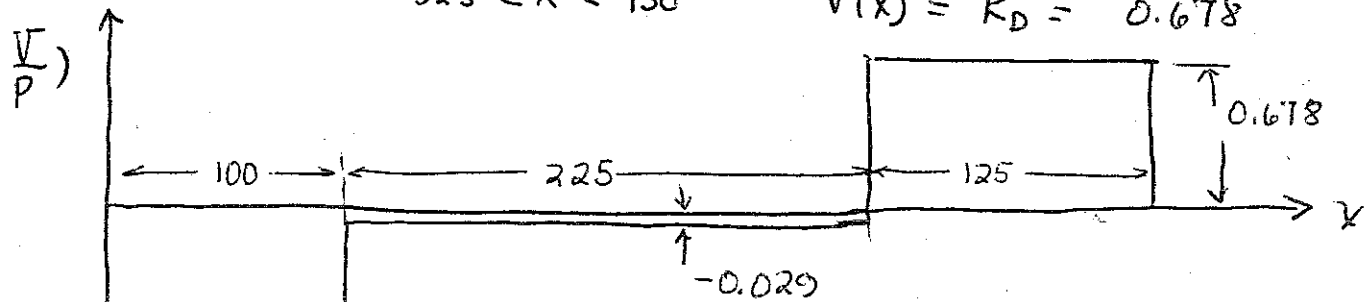
$$V(x) = -R_A = -0.736 P$$

$$100 < x < 325$$

$$V(x) = -R_A + \frac{P}{\sqrt{2}} = -0.029 P$$

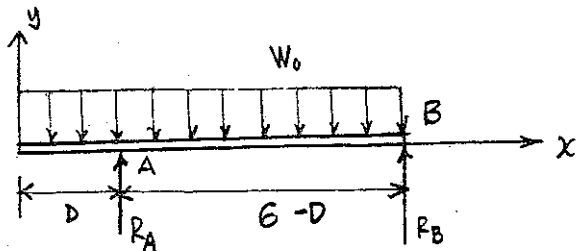
$$325 < x < 450$$

$$V(x) = R_D = 0.678 P$$



OBTAIN M AS SUPERPOSITION OF M DUE TO VERTICAL SHEAR AND HORIZONTAL FORCES

(3-17)



Moment equilibrium about B

$$\sum M_B = 0 \quad + \curvearrowright$$

$$\frac{WL^2}{2} - (6-D)R_A = 0 \quad L = 6$$

$$R_A = \frac{W_0 L^2}{2(6-D)}$$

loading $q(x) = -W_0 \langle x \rangle^0 + R_A \langle x-D \rangle^1$

$$\frac{dV}{dx} + q = 0 \rightarrow V = -R_A \langle x-D \rangle^0 + W_0 \langle x \rangle^1 + C_1$$

But $V(0) = 0 \quad C_1 = 0$

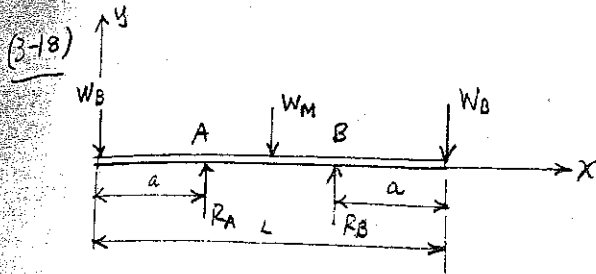
$$\frac{dM}{dx} + V = 0 \rightarrow M = -\frac{W_0}{2} \langle x \rangle^2 + R_A \langle x-D \rangle^1 + C_2$$

$M(0) = 0 \quad C_2 = 0$

WE ALSO HAVE THE BC $M(1.8) = 0$

$$\frac{W_0 \langle 1.8 \rangle^2}{2} = \frac{W_0 (6)^2}{2(6-D)} (1.8-D)$$

$$D = 1.385 \text{ m}$$



(a) Apparently he is not minimizing the bending moment by putting the bricks at the end. The bending moment would be less if he puts the bricks right above the supports.

$$\sum F_y = 0$$

$$R_A + R_B = W_M + 2W_B$$

$$\sum M_A = 0 \quad + \curvearrowleft$$

$$W_B a - W_M \left(\frac{L}{2} - a \right) + R_B (L - 2a) - W_B (L - a) = 0$$

$$R_B (L - 2a) = W_B (L - 2a) + W_M \left(\frac{L}{2} - a \right) = \left(W_B + \frac{W_M}{2} \right) (L - 2a)$$

$$\therefore R_B = W_B + \frac{W_M}{2} \rightarrow R_A = W_M + 2W_B - W_B - \frac{W_M}{2} = \frac{W_M}{2} + W_B$$

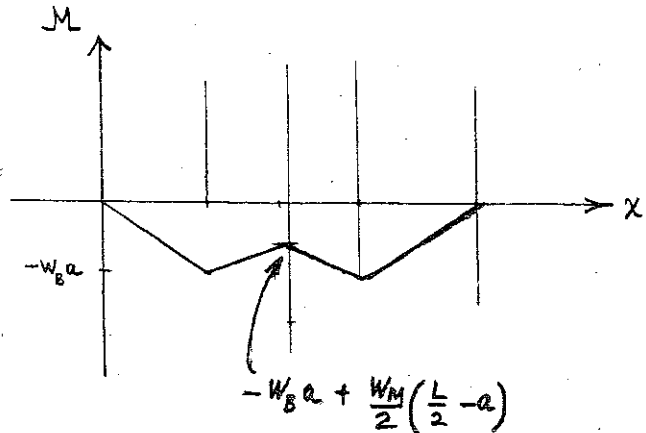
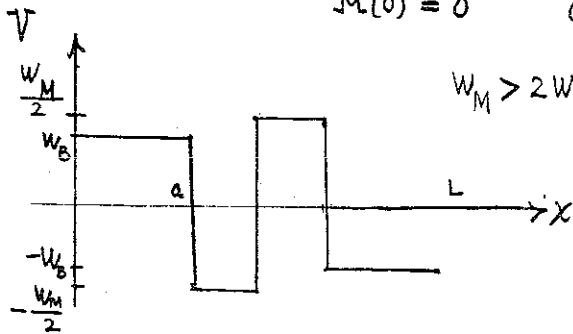
$$q(x) = -W_B \langle x \rangle_1 + R_A \langle x - a \rangle_1 - W_M \langle x - \frac{L}{2} \rangle_1 + R_B \langle x - L + 2a \rangle_1$$

$$\frac{dV}{dx} + q = 0 \rightarrow V = +W_B \langle x \rangle^0 - R_A \langle x - a \rangle^0 + W_M \langle x - \frac{L}{2} \rangle^0 - R_B \langle x - L + 2a \rangle^0 + C_1$$

$$V(0) = 0 \therefore C_1 = 0$$

$$\frac{dM}{dx} + V = 0 \rightarrow M = -W_B \langle x \rangle' + R_A \langle x - a \rangle' - W_M \langle x - \frac{L}{2} \rangle' + R_B \langle x - L + 2a \rangle' + C_2$$

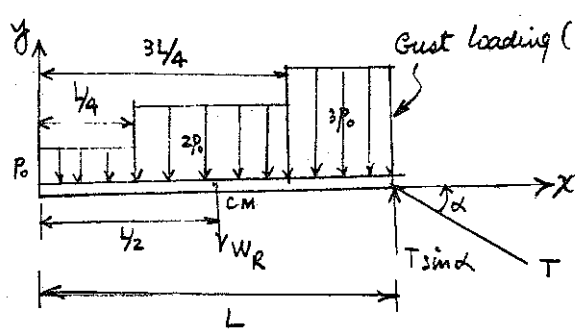
$$M(0) = 0 \quad C_2 = 0$$



To minimize the max. bending moment,

$$-W_B a = -\left[-W_B a + \frac{W_M}{2} \left(\frac{L}{2} - a \right) \right] \rightarrow W_B = \frac{W_M}{4a} \left(\frac{L}{2} - a \right)$$

(3-19)



Gust loading (load/unit length)

• Moment about C.M. $\sum \tau = 0$

+ (moment due P_0 loading) - (moment due to $3P_0$)

+ $T \sin \alpha \frac{L}{2} = 0 \dots \dots (1)$

moment due to P_0 loading

$$M_1 = \int_{L/4}^{L/2} P_0 x dx = \frac{P_0}{2} \left[\left(\frac{L}{2} \right)^2 - \left(\frac{L}{4} \right)^2 \right]$$

moment due to $3P_0$ loading

$$M_2 = 3 \int_{L/4}^{L/2} P_0 x dx = \frac{3P_0}{2} \left[\left(\frac{L}{2} \right)^2 - \left(\frac{L}{4} \right)^2 \right]$$

(1) becomes

$$-\frac{P_0}{2} \left[\left(\frac{L}{2} \right)^2 - \left(\frac{L}{4} \right)^2 \right] + \frac{3P_0}{2} \left[\left(\frac{L}{2} \right)^2 - \left(\frac{L}{4} \right)^2 \right] = \frac{TL}{2} \sin \alpha$$
$$P_0 \left[\frac{3L^2}{16} \right] = \frac{TL}{2} \sin \alpha$$

$$T = \frac{3}{8} \frac{P_0 L}{\sin \alpha}$$

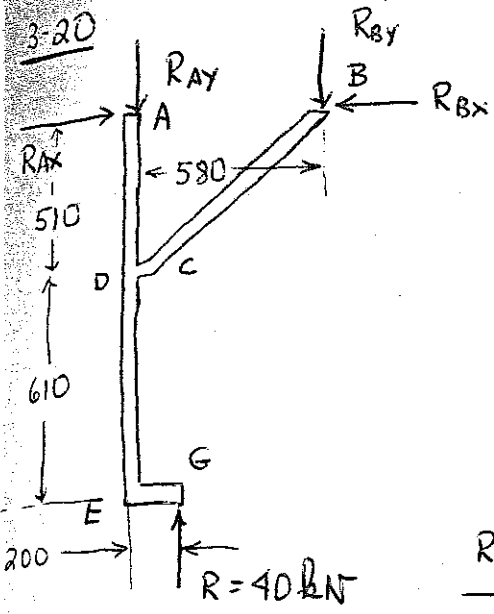
(b) $f(x) = -P_0 \langle x \rangle^0 - P_0 \langle x - \frac{L}{4} \rangle^0 - P_0 \langle x - \frac{3L}{4} \rangle^0$

$$V = - \int_{-\infty}^x f(x) dx = +P_0 \langle x \rangle^1 + P_0 \langle x - \frac{L}{4} \rangle^1 + P_0 \langle x - \frac{3L}{4} \rangle^1$$

$$M = - \int_{-\infty}^x V(x) dx = -\frac{P_0}{2} \langle x \rangle^2 - \frac{P_0}{2} \langle x - \frac{L}{4} \rangle^2 - P_0 \langle x - \frac{3L}{4} \rangle^2$$

∴ $V\left(\frac{L}{4}\right) = P_0 \frac{L}{4}$; $V\left(\frac{3L}{4}\right) = P_0 \frac{3L}{4} + P_0 \frac{L}{2} = \frac{5P_0 L}{4}$

$M\left(\frac{L}{4}\right) = -\frac{P_0}{16} L^2$; $M\left(\frac{3L}{4}\right) = -\frac{13}{32} P_0 L^2$



$$\sum F_y = 0$$

$$R_{Ay} + R_{By} = R$$

$$\sum F_x = 0$$

$$R_{Ax} = R_{Bx}$$

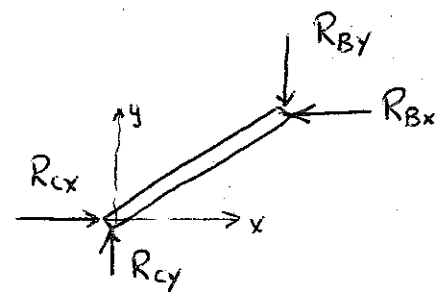
TAKING MOMENTS ABOUT A

$$\sum M_A = 0 \Rightarrow$$

$$R \cdot 200 - R_{By} \cdot 580 = 0$$

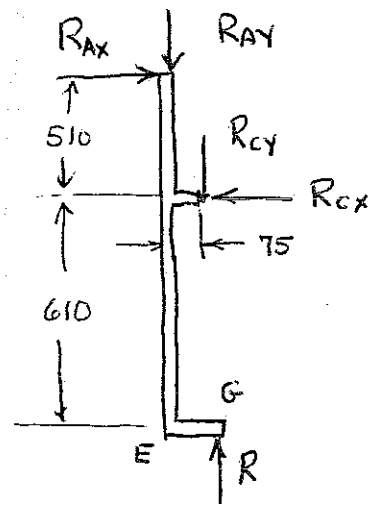
$$R_{By} = 13.8 \text{ kN}$$

$$R_{Ay} = 26.2 \text{ kN}$$



$$\sum F_y = 0 \Rightarrow R_{By} = R_{Cy}$$

$$\sum F_x = 0 \Rightarrow R_{Cx} = R_{Bx}$$



TAKING MOMENTS ABOUT A

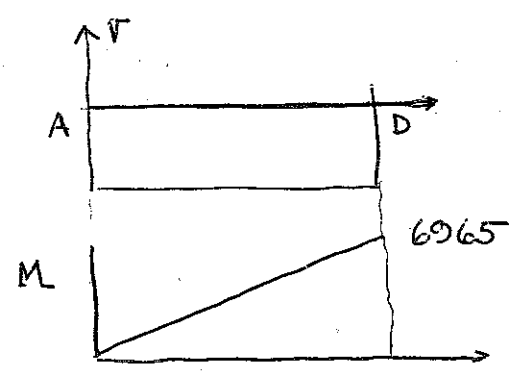
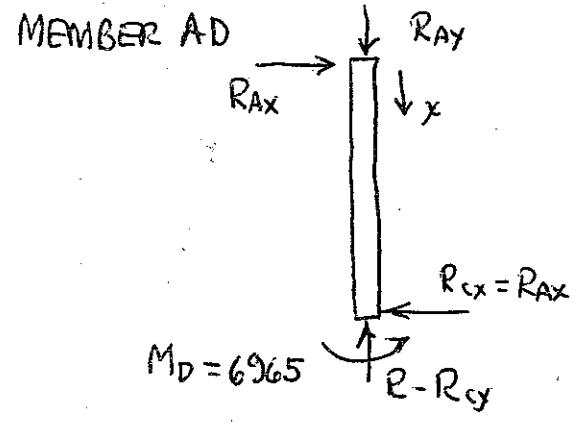
$$\sum M_A = 0 \Rightarrow$$

$$200 R - R_{Cy} \cdot 75 - R_{Cy} \cdot 510 = 0$$

$$R_{Cy} = 13.66 \text{ kN}$$

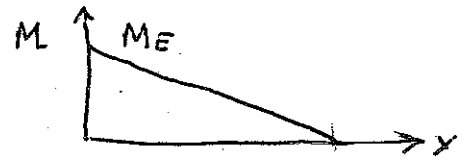
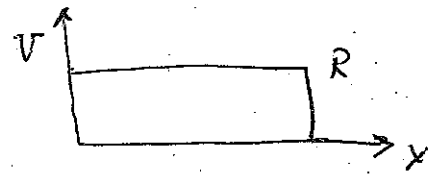
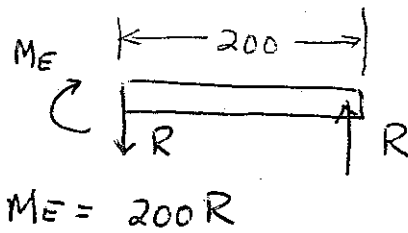
$\therefore$ THE REACTIONS ARE A: $R_{Ax} = 13.66 \text{ kN}$
 $R_{Ay} = 26.2 \text{ kN}$
 B: $R_{Bx} = 13.66 \text{ kN}$
 $R_{By} = 13.8 \text{ kN}$
 C: $R_{Cx} = 13.66 \text{ kN}$
 $R_{Cy} = 13.8 \text{ kN}$

V & M DISTRIBUTIONS IN MEMBERS:

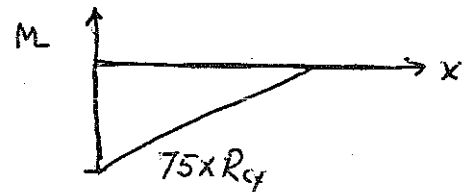
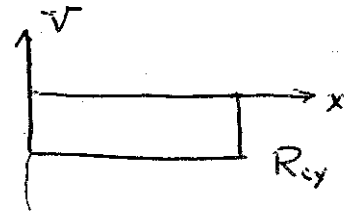
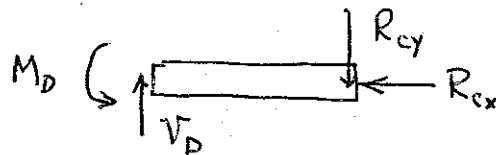


3-20 (CONT)

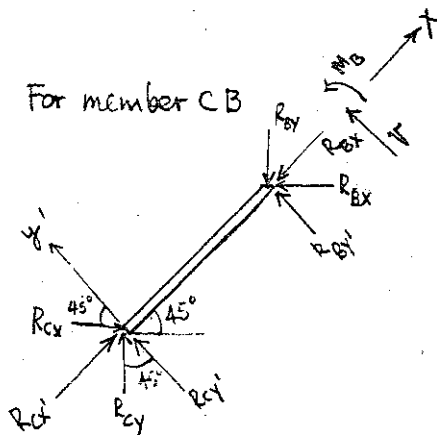
MEMBER EG:



MEMBER DC



SIMILARLY FOR DE

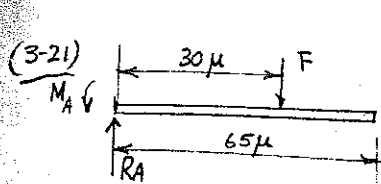


$$R_{Bx'} = R_{Cx'} \quad ; \quad R_{By'} = R_{Cy'}$$

$$R_{Cx'} = R_{cx} \sin 45^\circ + R_{cy} \cos 45^\circ = \frac{R_{cx}}{\sqrt{2}} + \frac{R_{cy}}{\sqrt{2}}$$

$$R_{Cy'} = R_{cy} \cos 45^\circ - R_{cx} \sin 45^\circ = 0$$

- (i) Compressive axial force
- (ii) There is no shear force acting along the member
Since $R_{Cy'} = R_{By'} = 0$
- (iii) There is no moment distribution.



(a) $\sum M = 0$

$$M_A = F \times 30 \times 10^{-4} \text{ cm.}$$

$$F = 2.2 \times 10^{-4} \text{ dyne}$$

$$M_A = 66 \times 10^{-8} \text{ dynes-cm.} = 6.6 \times 10^{-7} \text{ dyne-cm.}$$

$$1 \mu\text{m} = 10^{-6} \text{ m}$$

$$1 \text{ DYNE} = 10^{-5} \text{ N}$$

(b) Let force on element of length be dF

$$dF = v \eta f(x) dx$$

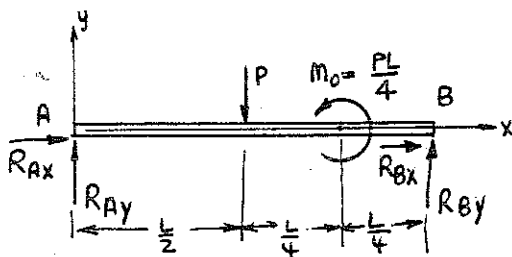
$$= \omega \eta x f(x) dx$$

$v = \text{angular velocity} = x \omega$ $\omega = \text{frequency}$
 $\eta = \text{viscosity}$

Moment due to this force

$$\int_0^L x dF = \int_0^L \omega \eta x^2 f(x) dx = \omega \eta \int_0^L x^2 f(x) dx$$

3.22



$$\sum (M_{\equiv})_A = 0 : R_{By}(L) + M_0 - P(\frac{L}{2}) = 0$$

$$\therefore R_{By} = \frac{P}{2} - \frac{M_0}{L} = \frac{P}{2} - \frac{P}{4} = \frac{P}{4}$$

$$\sum F_y = 0 : R_{Ay} + R_{By} - P = 0, \therefore R_{Ay} = \frac{3P}{4}$$

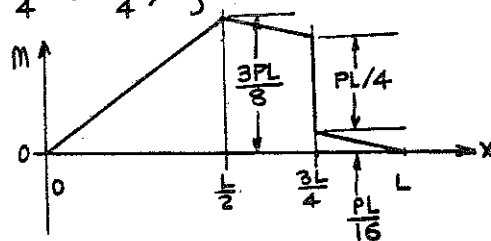
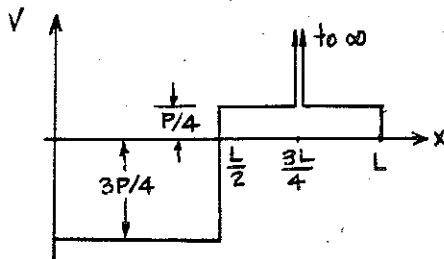
R_{Ax} and R_{Bx} cannot be determined but they do not affect the V, M distributions if deflections are small.

$$q(x) = \left\{ \frac{3P}{4} \langle x \rangle_1 - P \langle x - \frac{L}{2} \rangle_1 - \frac{PL}{4} \langle x - \frac{3L}{4} \rangle_2 + \frac{P}{4} \langle x - L \rangle_1 \right\}$$

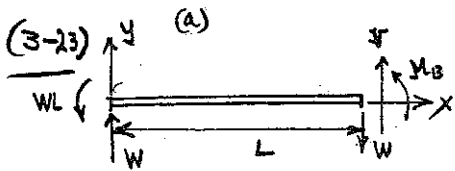
$$V(x) = -\int_{-\infty}^x q dx = \left\{ -\frac{3P}{4} \langle x \rangle^0 + P \langle x - \frac{L}{2} \rangle^0 + \frac{PL}{4} \langle x - \frac{3L}{4} \rangle_1 \right\}$$

(This term is not really necessary and we will drop it.)

$$M(x) = -\int_{-\infty}^x V dx = \left\{ \frac{3P}{4} \langle x \rangle^1 - P \langle x - \frac{L}{2} \rangle^1 - \frac{PL}{4} \langle x - \frac{3L}{4} \rangle^0 \right\}$$



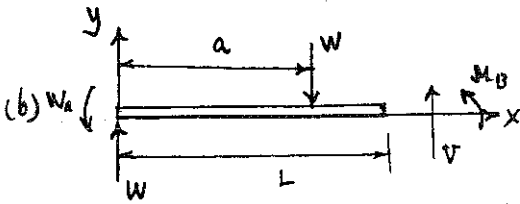
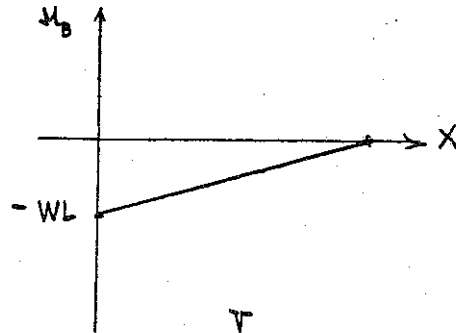
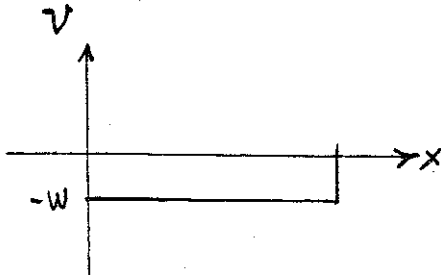
(3-23) (a)



$$q(x) = -WL \langle x \rangle_{-2}^0 + W \langle x \rangle_{-1}^1 - W \langle x-L \rangle_{-1}^0$$

$$V(x) = WL \langle x \rangle_{-1}^0 - W \langle x \rangle_{-1}^1 + W \langle x-L \rangle_{-1}^0$$

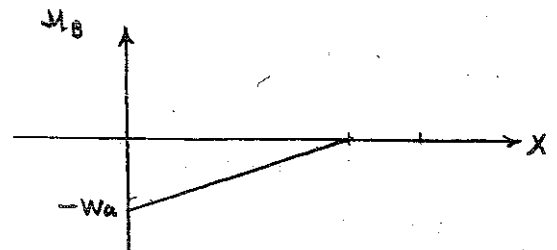
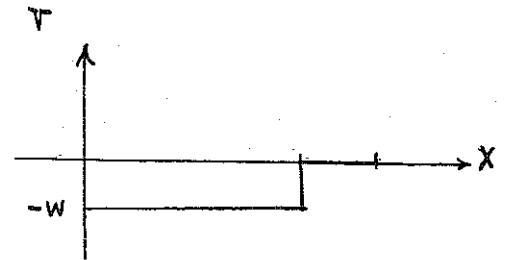
$$M_B(x) = -WL \langle x \rangle_{-1}^0 + W \langle x \rangle_{-1}^1 - W \langle x-L \rangle_{-1}^0$$



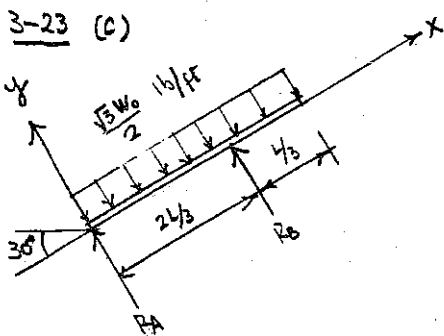
$$q(x) = -W_a \langle x \rangle_{-2}^0 + W \langle x \rangle_{-1}^1 - W \langle x-a \rangle_{-1}^0$$

$$V(x) = W_a \langle x \rangle_{-1}^0 - W \langle x \rangle_{-1}^1 + W \langle x-a \rangle_{-1}^0$$

$$M_B(x) = -W_a \langle x \rangle_{-1}^0 + W \langle x \rangle_{-1}^1 - W \langle x-a \rangle_{-1}^0$$



3-23 (c)

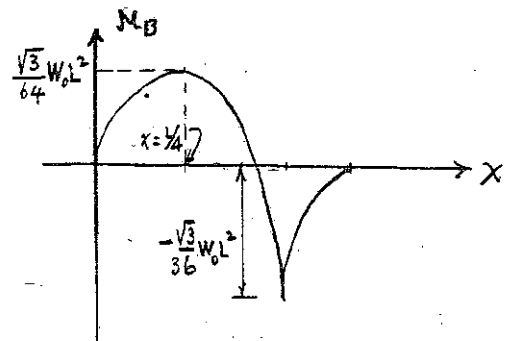
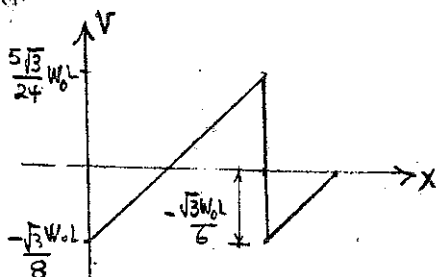


$$R_A = \frac{\sqrt{3}}{8} W_0 L \quad R_B = \frac{3\sqrt{3}}{8} W_0 L \quad (\text{From Prob. 3-3})$$

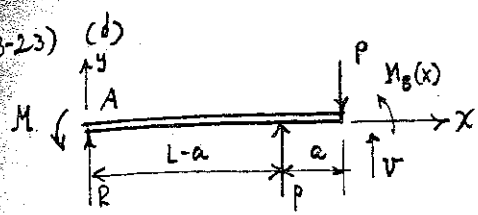
$$q(x) = R_A \langle x \rangle_{-1}^0 - W_0 \frac{\sqrt{3}}{2} \langle x \rangle_{-1}^1 + R_B \langle x - \frac{2L}{3} \rangle_{-1}^0$$

$$V(x) = -R_A \langle x \rangle_{-1}^0 + \frac{\sqrt{3} W_0}{2} \langle x \rangle_{-1}^1 - R_B \langle x - \frac{2L}{3} \rangle_{-1}^0$$

$$M_B(x) = R_A \langle x \rangle_{-1}^0 - \frac{\sqrt{3} W_0}{2} \langle x \rangle_{-1}^1 + R_B \langle x - \frac{2L}{3} \rangle_{-1}^0$$



(3-23)



$$\sum M_A = 0 \quad +\curvearrowright$$

$$M + P(L-a) - PL = 0$$

$$M = Pa$$

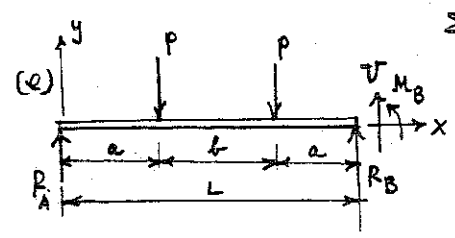
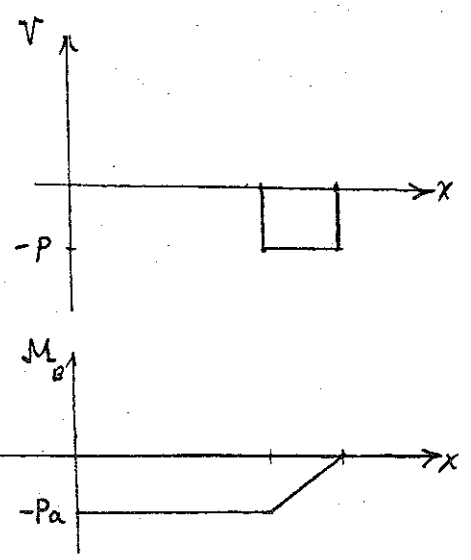
$$\sum F_y = 0$$

$$R = 0$$

$$q(x) = P\langle x-L+a \rangle_{-1} - M\langle x \rangle_2 = P\langle x-L+a \rangle_{-1} - Pa\langle x \rangle_2$$

$$V(x) = -\int_{-\infty}^x q(x)dx = Pa\langle x \rangle_{-1} - P\langle x-L+a \rangle^0$$

$$M(x) = -\int_{-\infty}^x V(x)dx = -Pa\langle x \rangle^0 + P\langle x-L+a \rangle^1$$



$$\sum F_y$$

$$R_A + R_B = 2P$$

$$\sum M_B = 0 \quad +\curvearrowright$$

$$-R_A L + P(a+b) + Pa = 0$$

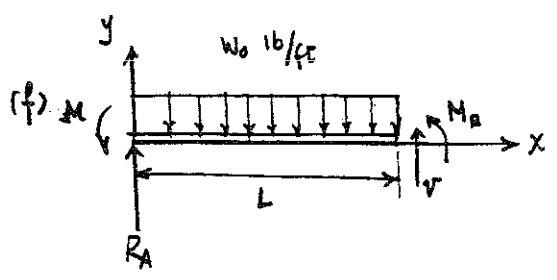
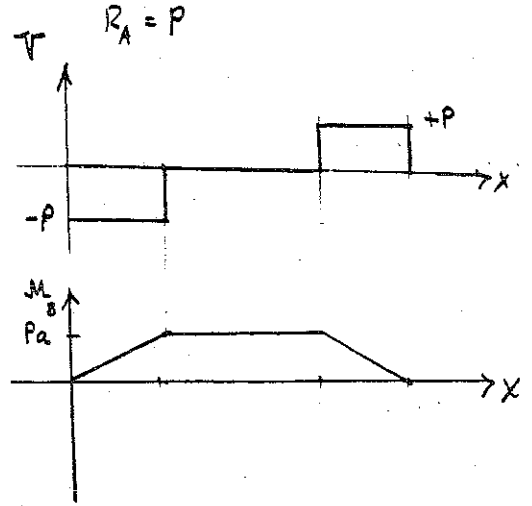
$$R_A = P$$

$$\therefore R_B = P$$

$$q(x) = P\langle x \rangle_{-1} - P\langle x-a \rangle_{-1} - P\langle x-a-b \rangle_{-1}$$

$$V(x) = -P\langle x \rangle^0 + P\langle x-a \rangle^0 + P\langle x-a-b \rangle^0$$

$$M(x) = -P\langle x \rangle^1 + P\langle x-a \rangle^1 + P\langle x-a-b \rangle^1$$



$$\sum F_y = 0$$

$$R_A = w_0 L$$

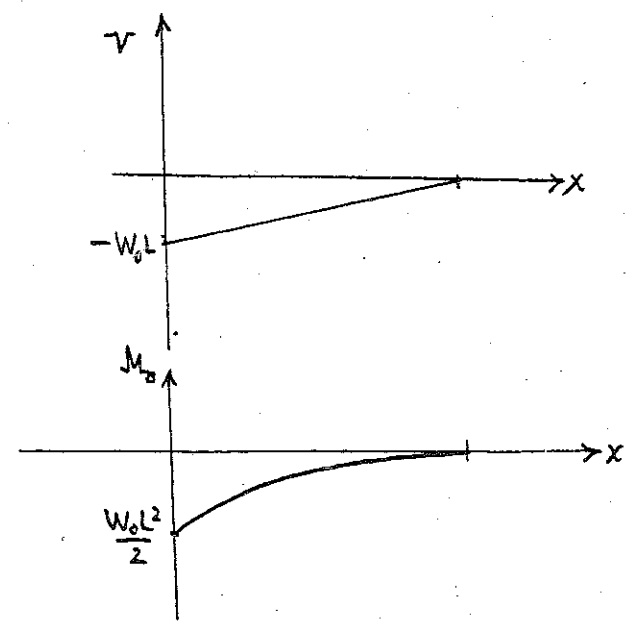
$$\sum M = 0 \quad +\curvearrowright$$

$$M = \frac{w_0 L^2}{2}$$

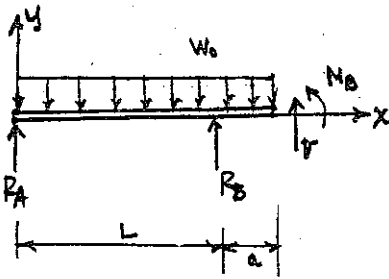
$$q(x) = -\frac{w_0 L^2}{2}\langle x \rangle_{-2} + w_0 L\langle x \rangle_{-1} - w_0\langle x \rangle^0$$

$$V(x) = \frac{w_0 L^2}{2}\langle x \rangle_{-1} - w_0 L\langle x \rangle^0 + w_0\langle x \rangle^1$$

$$M(x) = -\frac{w_0 L^2}{2}\langle x \rangle^0 + w_0 L\langle x \rangle^1 - \frac{w_0\langle x \rangle^2}{2}$$



(3-23)
(g)



$$\sum F_y = 0$$

$$R_A + R_B = w_0(L + a)$$

$$R_A = \frac{w_0}{2} \frac{L^2 - a^2}{L}$$

$$\sum M_A = 0 + \curvearrowright$$

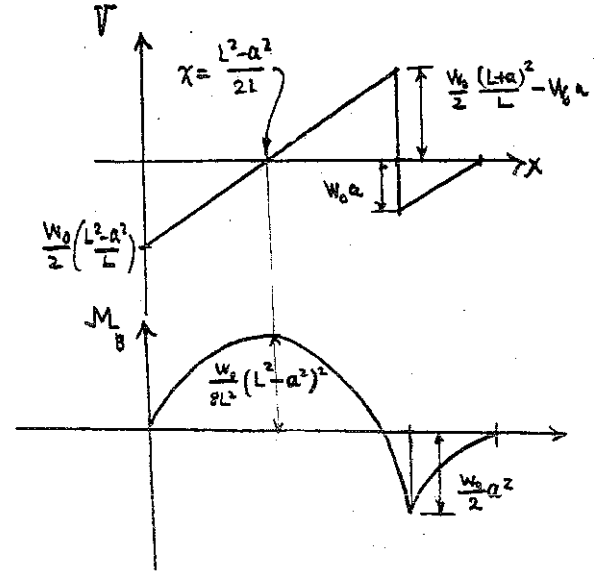
$$R_B L - w_0 \frac{(L + a)^2}{2} = 0$$

$$R_B = \frac{w_0}{2} \frac{(L + a)^2}{L}$$

$$q(x) = -w_0 \langle x \rangle^0 + \frac{w_0}{2} \frac{L^2 - a^2}{L} \langle x \rangle^{-1} + \frac{w_0 (L + a)^2}{2} \langle x - L \rangle^{-1}$$

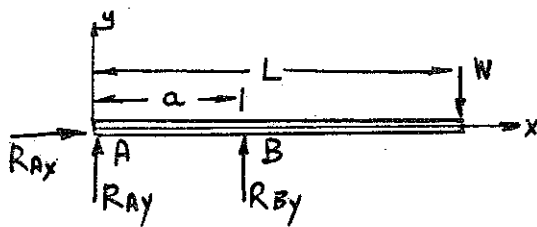
$$V(x) = w_0 \langle x \rangle^1 - \frac{w_0}{2} \frac{L^2 - a^2}{L} \langle x \rangle^0 - \frac{w_0 (L + a)^2}{2} \langle x - L \rangle^0$$

$$M(x) = -\frac{w_0}{2} \langle x \rangle^2 + \frac{w_0}{2} \frac{L^2 - a^2}{L} \langle x \rangle^1 + \frac{w_0 (L + a)^2}{2} \langle x - L \rangle^1$$



(h) See 3-8 solution

3.24



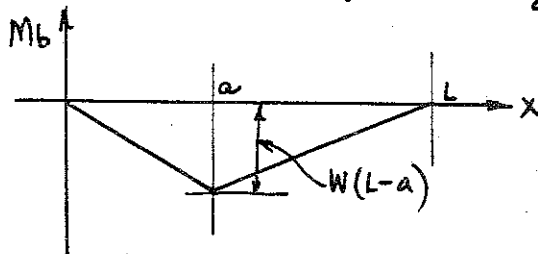
Assume : zero moment at A;
zero horizontal force at B;
negligible deflections.

$$\sum (M_z)_A = 0 : R_{By}(a) - WL = 0 \quad \therefore R_{By} = \frac{WL}{a}$$

$$\sum F_y = 0 : R_{Ay} + R_{By} - W = 0 \quad \therefore R_{Ay} = W(1 - \frac{L}{a})$$

Loading, $q(x) = W(1 - \frac{L}{a})\langle x \rangle_{-1} + \frac{WL}{a}\langle x-a \rangle_{-1}$

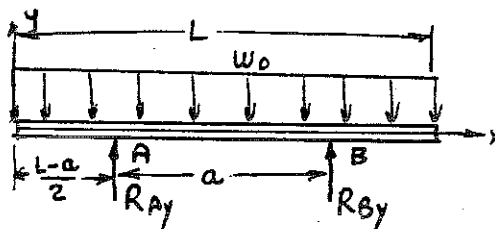
$$\therefore M_b(x) = W(1 - \frac{L}{a})\langle x \rangle' + \frac{WL}{a}\langle x-a \rangle'$$



For constant $(M_b)_{max}$, therefore

$$W(L-a) = \text{Constant}$$

3.25



Assume uniform loading w_0 .
Overall equilibrium :

$$\sum (M_z)_A = 0 : R_{By}(a) - w_0 L(\frac{a}{2}) = 0$$

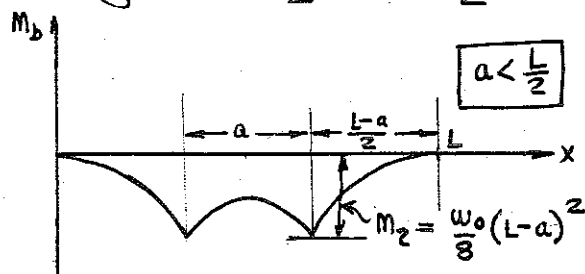
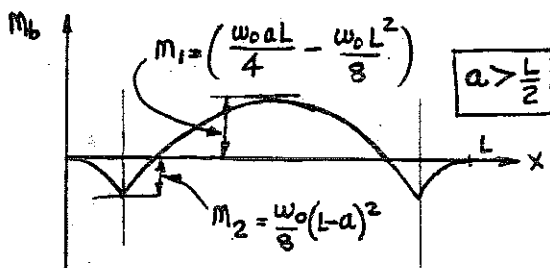
$$\therefore R_{By} = \frac{w_0 L}{2}$$

$$\sum (F_y) = 0 : R_{Ay} + R_{By} - w_0 L = 0 \quad \therefore R_{Ay} = \frac{w_0 L}{2}$$

Loading, $q(x) = -w_0\langle x \rangle'' + \frac{w_0 L}{2}\langle x - \frac{L}{2} + \frac{a}{2} \rangle_{-1} + \frac{w_0 L}{2}\langle x - \frac{L}{2} - \frac{a}{2} \rangle_{-1}$

$$\therefore M_b(x) = -\frac{w_0}{2}\langle x \rangle'' + \frac{w_0 L}{2}\langle x - \frac{L}{2} + \frac{a}{2} \rangle' + \frac{w_0 L}{2}\langle x - \frac{L}{2} - \frac{a}{2} \rangle'$$

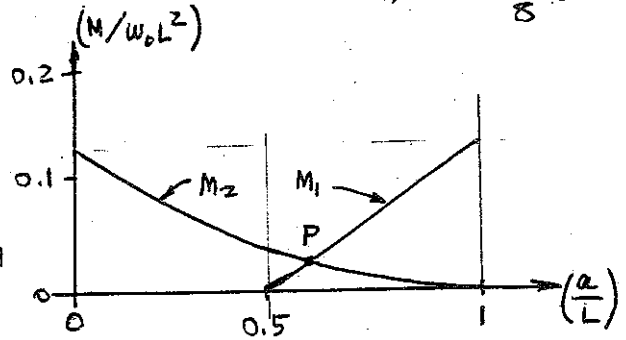
There are two cases according as $a > \frac{L}{2}$ or $a < \frac{L}{2}$:-



when $a > L/2$ there are two extreme values for M_b ,
 $M_1 = \left(\frac{w_0 L a}{4} - \frac{w_0 L^2}{8} \right)$ and $M_2 = \frac{w_0}{8} (L-a)^2$

When $a < L/2$ there is only one extreme value, $M_2 = \frac{w_0}{8} (L-a)^2$.

M_1 and M_2 are plotted as functions of (a/L) in the figure alongside. As can be seen, minimum bending moment occurs when $M_1 = M_2$.



Thus: $\frac{w_0 L a}{4} - \frac{w_0 L^2}{8} = \frac{w_0}{8} (L-a)^2$; giving $\left(\frac{a}{L} \right) = 0.586$

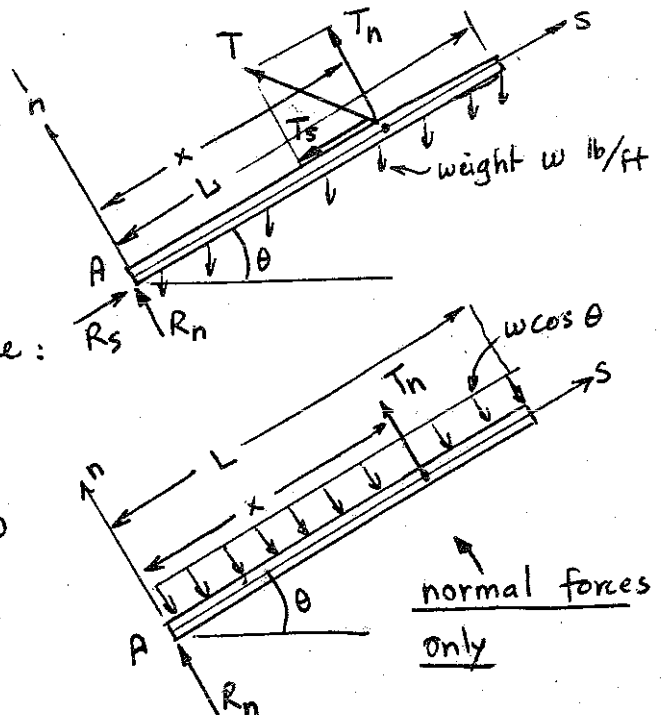
3.26 If the flagpole does not bend to an appreciable extent, then bending moment can only be caused by forces normal to the flagpole. These are:

- 1) R_n
- 2) T_n
- 3) normal component of w
 $= w \cos \theta$.

$$\underline{\sum (M_{\perp})_A = 0} : T_n x = w \cos \theta \frac{L^2}{2}$$

$$\therefore T_n = \frac{w L^2}{2x} \cos \theta$$

$$\underline{\sum (F_n) = 0} : R_n = (w \cos \theta) L - \frac{w L^2}{2x} \cos \theta$$



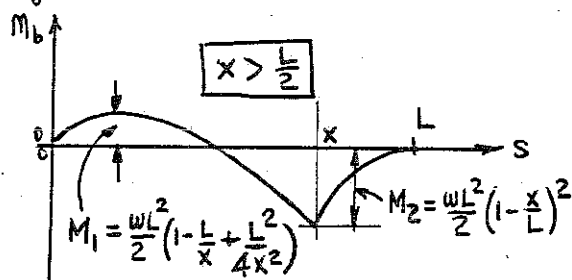
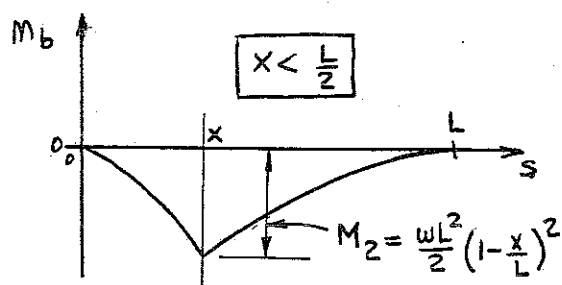
Since all the forces are proportional to $\cos \theta$, the bending moment will also be proportional to it and the worst case is when $\theta = 0$, $\cos \theta = 1$. We will place $\theta = 0$ in the following analysis.

$$q(s) = R_h \langle s \rangle_1 + T_h \langle s-x \rangle_1 - w \langle s \rangle^0$$

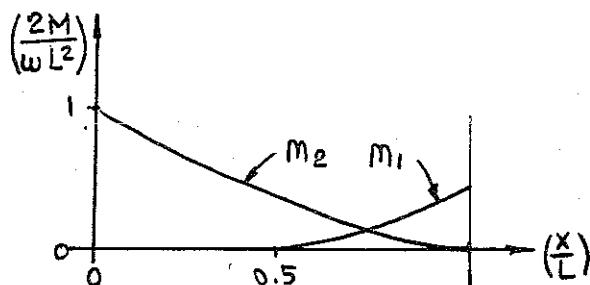
$$= \left(\omega L - \frac{\omega L^2}{2x} \right) \langle s \rangle_1 + \frac{\omega L^2}{2x} \langle s-x \rangle_1 - w \langle s \rangle^0$$

$$M_b(s) = \left(\omega L - \frac{\omega L^2}{2x} \right) \langle s \rangle^1 + \frac{\omega L^2}{2x} \langle s-x \rangle^1 - \frac{w}{2} \langle s \rangle^2$$

This equation has two maxima (for the absolute value of M_b) for $x > \frac{L}{2}$ and only one for $x < \frac{L}{2}$:-



There is always a max. value ($= M_2$) at $s=x$; also when $x > \frac{L}{2}$ there is another max. at $s < x$. We will plot M_1 and M_2 as functions of $\left(\frac{x}{L}\right)$



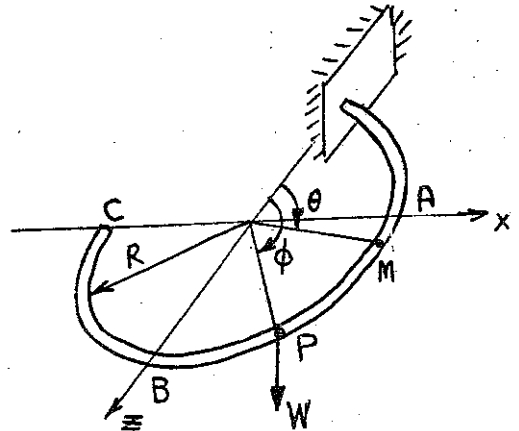
Minimum M_b occurs at the point of intersection of M_1 and M_2 curves.

For $M_1 = M_2$: $\left(1 - \frac{L}{x} + \frac{L^2}{4x^2}\right) = \left(1 - \frac{x}{L}\right)^2$, leading to the quartic equation : $4\left(\frac{x}{L}\right)^4 - 8\left(\frac{x}{L}\right)^3 + 4\left(\frac{x}{L}\right) - 1 = 0$. Solving, $\frac{x}{L} = 0.7$

or $x = 52.5 \text{ ft.}$

3.27

Consider the general case where a load W is applied at point P (angle ϕ) and we wish to find the bending and twisting moments for any point M at angle θ



Clearly, for $\theta > \phi$, $M_b = M_t = 0$

For $\theta < \phi$: (See projection)

Projection of free body:

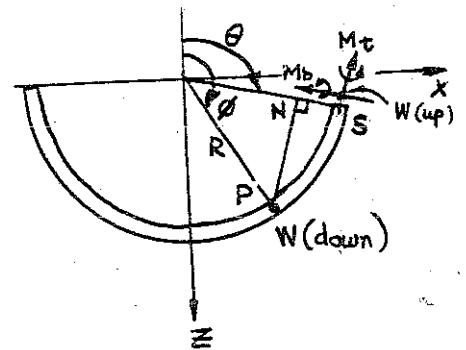
$$M_b = W(PN)$$

$$M_t = W(NS)$$

$$PN = R \sin(\phi - \theta)$$

$$NS = R [1 - \cos(\phi - \theta)]$$

$$\therefore \begin{cases} M_b = WR \sin(\phi - \theta) \\ M_t = WR [1 - \cos(\phi - \theta)] \end{cases} \quad \theta \leq \phi$$



1) Load at A, $\phi = 90^\circ$

$$M_b = WR \cos \theta \quad ; (M_b)_{\max} = WR \text{ at } \theta = 0^\circ$$

$$M_t = WR [1 - \sin \theta] \quad ; (M_t)_{\max} = WR \text{ at } \theta = 0^\circ$$

2) Load at B, $\phi = 180^\circ$

$$M_b = WR \sin \theta \quad ; (M_b)_{\max} = WR \text{ at } \theta = 90^\circ$$

$$M_t = WR [1 + \cos \theta] \quad ; (M_t)_{\max} = 2WR \text{ at } \theta = 0^\circ$$

3) Load at C, $\phi = 270^\circ$

$$M_b = -WR \cos \theta \quad ; (M_b)_{\max} = WR \text{ at } \theta = 0^\circ, 180^\circ$$

$$M_t = WR [1 + \sin \theta] \quad ; (M_t)_{\max} = 2WR \text{ at } \theta = 90^\circ$$

3.28 In s, n coordinates

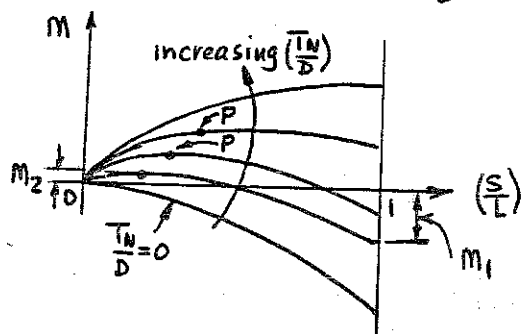
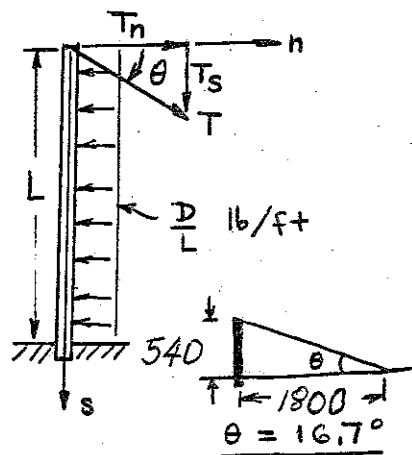
$$q(s) = T_n \langle s \rangle_1 - \frac{D}{L} \langle s \rangle^0$$

$$M_b(s) = T_n \langle s \rangle^1 - \frac{D}{2L} \langle s \rangle^2$$

$$= T_n s - \frac{D}{2L} s^2$$

$$= \frac{DL}{2} \left[-\left(\frac{s}{L}\right)^2 + 2\left(\frac{T_n}{D}\right)\left(\frac{s}{L}\right) \right]$$

The shape of this curve depends on the value of $\left(\frac{T_n}{D}\right)$.



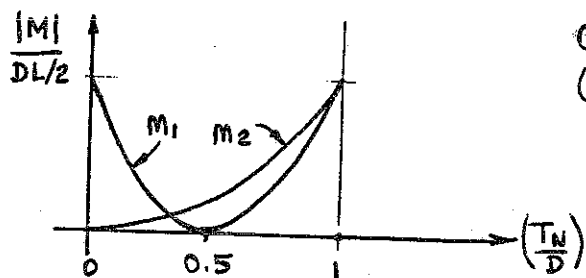
The max. will occur either at $\left(\frac{s}{L}\right)=1$ or at points such as P when these exist. We will denote the moment at $\frac{s}{L}=1$ by M_1 and that at P by M_2 .

At P, $\frac{dM}{ds} = 0 \Rightarrow -2\left(\frac{s}{L}\right) + 2\left(\frac{T_n}{D}\right) = 0$ i.e. $\left(\frac{s}{L}\right) = \left(\frac{T_n}{D}\right)$

$\therefore$ P will exist if $0 < \left(\frac{T_n}{D}\right) \leq 1$, and in that case,

$$M_2 = \frac{DL}{2} \left[-\left(\frac{T_n}{D}\right)^2 + 2\left(\frac{T_n}{D}\right)\left(\frac{T_n}{D}\right) \right] = \frac{DL}{2} \left(\frac{T_n}{D}\right)^2$$

$M_1 = \frac{DL}{2} \left[-1 + 2\left(\frac{T_n}{D}\right) \right]$. We will plot M_1 and M_2 as functions of $\left(\frac{T_n}{D}\right)$ to locate the optimum $\left(\frac{T_n}{D}\right)$.



Optimum value is when $M_1 = M_2$ (absolute values)

$$\text{i.e. } \left(\frac{T_n}{D}\right)^2 = \left[1 - 2\left(\frac{T_n}{D}\right) \right]$$

$$\left(\frac{T_n}{D}\right) = \sqrt{2} - 1 = 0.414$$

$$T_n = T \cos \theta = 0.958 T$$

$$\therefore \left(\frac{T}{D}\right) = \frac{0.414}{0.958} = \underline{\underline{0.432}}$$

3.29 Consider the free body AB formed by making a cut at A so that the cross section at A is normal to the wire axis.

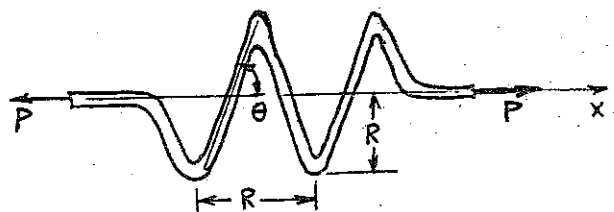
We define a local orthogonal system as follows:

n : normal to the cross section.

r : tangential to the section and $\perp$ to the x axis

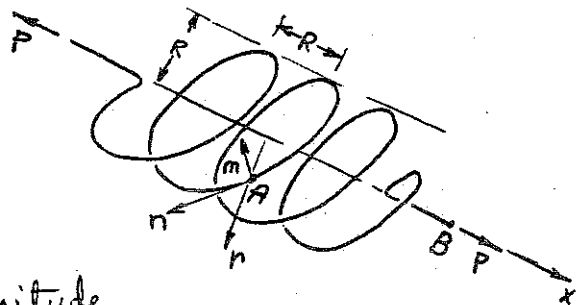
m : $\perp$ to n and r

The force P will have a moment at A equal in magnitude to PR and with direction perpendicular to r and x . We resolve this vector moment into n, r, m components to obtain M_b and M_t .



$$\text{pitch} = 2\pi(\text{radius}) \cot \theta$$

$$\therefore \cot \theta = \frac{R}{2\pi R} = \frac{1}{2\pi}; \quad \theta = 80.9^\circ$$



Thus:

$$M_r = 0$$

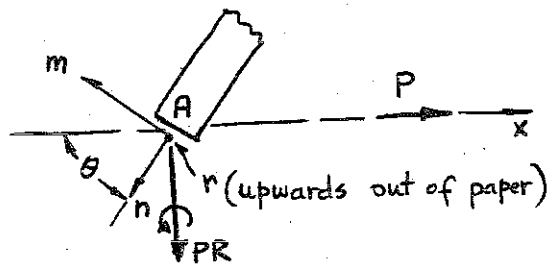
$$M_n = M_t = PR \cos(90 - \theta) = PR \sin \theta$$

$$M_m = M_b = PR \sin(90 - \theta) = PR \cos \theta$$

And these are independent of the position of A.

$$\therefore (M_b)_{\max} = PR \cos(80.9^\circ) = \underline{0.158 PR}$$

$$(M_t)_{\max} = PR \sin(80.9^\circ) = \underline{0.985 PR}$$



3.30 Equilibrium for free bodies :-

(x-y is the plane of the 18" and 24" legs).

(i)

$$\sum M_x = 0 : -M_{bx} + \frac{500}{\sqrt{2}} (18-y) = 0$$

$$\therefore M_{bx} = \frac{500}{\sqrt{2}} (18-y)$$

$$\sum M_y = 0 : -M_t - \frac{500}{\sqrt{2}} (6) = 0$$

$$\therefore M_t = -\frac{3000}{\sqrt{2}} \text{ lb-in}$$

$$\sum M_z = 0 : -M_{bz} - \frac{500}{\sqrt{2}} (6) = 0$$

$$\therefore M_{bz} = -\frac{3000}{\sqrt{2}}$$

max. M_b at B ($y=0$)

$$M_{bx} = \frac{9000}{\sqrt{2}} \text{ in-lb}, M_{bz} = -\frac{3000}{\sqrt{2}}$$

$$(M_b)_{\max} = \sqrt{M_{bx}^2 + M_{bz}^2} = \underline{6700 \text{ lb-in}}$$

$$(M_t)_{\max} = \underline{2120 \text{ lb-in.}}$$

(ii)

$$\sum M_x = 0 : -M_t + \frac{500}{\sqrt{2}} (18) = 0$$

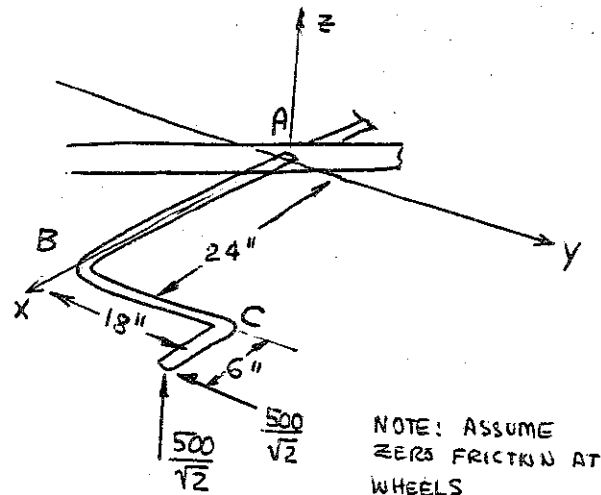
$$\therefore M_t = \frac{9000}{\sqrt{2}} \text{ lb-in.}$$

$$\sum M_y = 0 : -M_{by} - \frac{500}{\sqrt{2}} [6 + (24-x)]$$

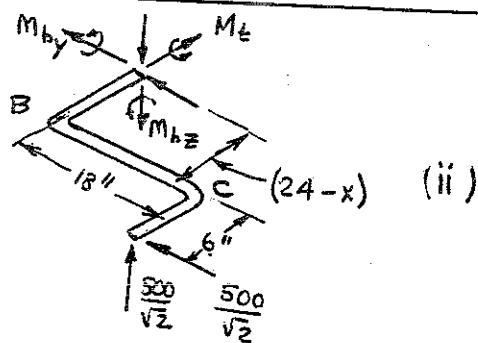
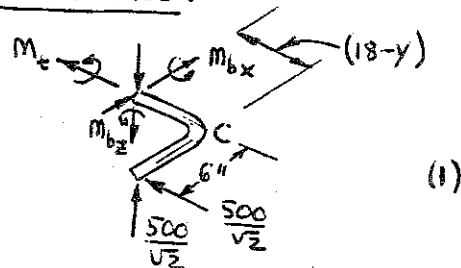
$$\therefore M_{by} = -\frac{500}{\sqrt{2}} (30-x)$$

$$\sum M_z = 0 : -M_{bz} - \frac{500}{\sqrt{2}} [6 + (24-x)] = 0$$

$$\therefore M_{bz} = -\frac{500}{\sqrt{2}} (30-x)$$



Free bodies:



max. M_b at A ($x=0$)

$$M_{by} = -\frac{15000}{\sqrt{2}} \text{ lb-in}$$

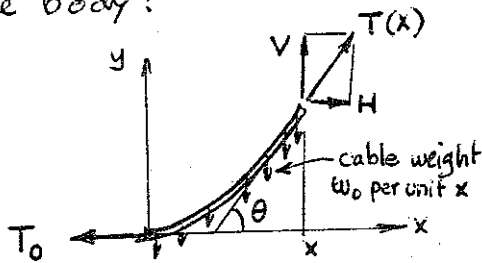
$$M_{bz} = -\frac{15000}{\sqrt{2}} \text{ lb-in.}$$

$$(M_b)_{\max} = \underline{15,000 \text{ lb-in}}$$

$$(M_t)_{\max} = \underline{6360 \text{ lb-in}}$$

3.31 The tension in the cable must everywhere be tangent to the cable (cable will not support any bending)

Free body:



At $x=0$ the tension $= T_0$ and is horizontal.

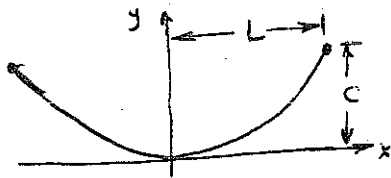
The weight is a downward force. Thus:

$$\sum F_x = 0 : H - T_0 = 0 \text{ ie } \underline{H = T_0}$$

$\therefore H = T_0$, independent of x , ie. $H = \text{constant}$.

Total load on cable from $x=0$ to $x=x$ is $w_0 x$.

$$\therefore \underline{\sum F_y = 0} : V = H \tan \theta = w_0 x ; \text{ ie } \tan \theta = \frac{dy}{dx} = \frac{w_0 x}{H}$$



$$y = \frac{w_0 x^2}{2H} + A ; \text{ at } x=0, y=0 ; \therefore \underline{A=0}$$

$$y = \frac{w_0 x^2}{2H} . \text{ at } x=L, y=C \text{ ie. } \underline{C = \frac{w_0 L^2}{2H}}$$

$$\therefore \underline{H = \frac{w_0 L^2}{2C}}$$

3.32 $\underline{\sum F_x = 0} : R_{BH} \sin \theta = 0$

$$\therefore \underline{R_{BH} = 0}$$

$$\underline{\sum F_y = 0} : R_{Ay} + R_{By} = 0 \text{ --- (1)}$$

$$\underline{\sum F_z = 0} : R_{Az} - R_{BH} \cos \theta = 0.$$

$$\therefore \underline{R_{Az} = 0}.$$

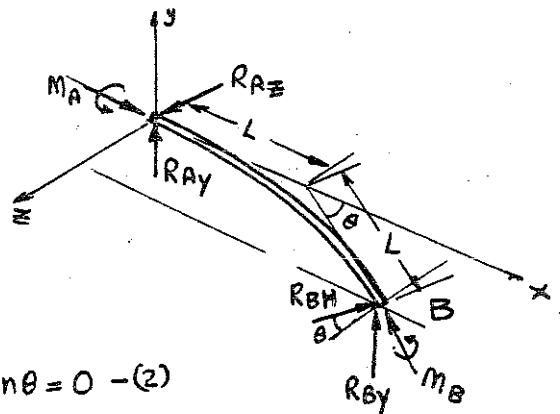
$$\underline{\sum (M_x)_A = 0} : M_A - M_B \cos \theta - R_{By} L \sin \theta = 0 \text{ --- (2)}$$

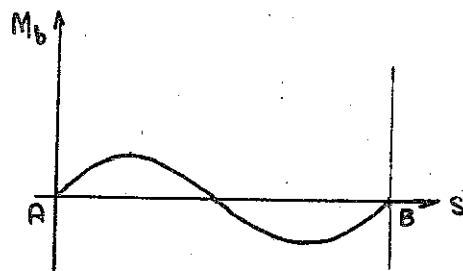
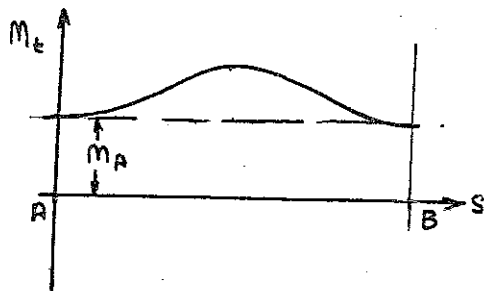
$$\underline{\sum (M_y)_A = 0} : \text{ identically satisfied}$$

$$\underline{\sum (M_z)_A = 0} : -M_B \sin \theta + R_{By} L (1 + \cos \theta) = 0 \text{ --- (3)}$$

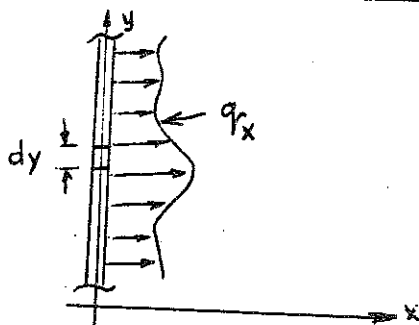
$$\text{Solving (1),(2),(3)} : \underline{R_{Ay} = -\frac{M_A}{L} \tan\left(\frac{\theta}{2}\right) = -R_{By}}$$

$$\text{and } \underline{M_B = M_A}$$

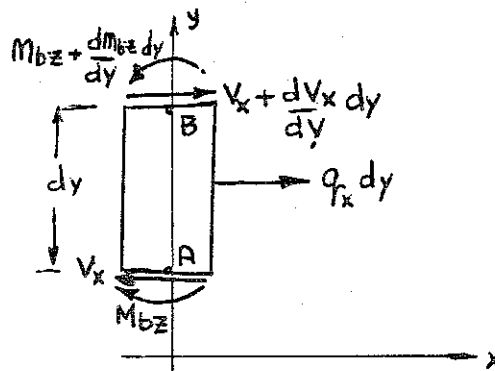




3.33



Free body diagram of slice of length dy



$$\sum F_x = 0 : V_x + \frac{dV_x}{dy} dy - V_x + q_x dy = 0$$

$$\therefore \boxed{\frac{dV_x}{dy} + q_x = 0}$$

$$\sum (M_z)_B = 0 : M_{Bz} + \frac{dM_{Bz}}{dy} dy - V_x dy + q_x dy \frac{dy}{2} - M_{Bz} = 0$$

$$\therefore \boxed{\frac{dM_{Bz}}{dy} - V_x = 0}$$

Corresponding to each combination we can write a direction triplet : (direction of member) (direction of loading) (direction of moment vector). [In the above example, this triplet is (y)(x)(z)]. If this triplet is ordered according to the right hand rule, equations corresponding to (a) apply. Otherwise the equations corresponding to (b) apply.

direction of member	loading	equation type	triplet
x	z	(b)	xzy
y	z	(a)	yzx
z	x	(a)	zxy
z	y	(b)	zyx

3.34

The stresses in the planks are symmetrical about the x axis. Thus there can be no shear stresses τ_{zy} at A and B in plank ① and similar points in the other planks. Now consider the y direction equilibrium of a plank. The y direction forces are:

- 1) linearly distributed water pressure
- 2) force exerted by the uprights (unknown distribution)

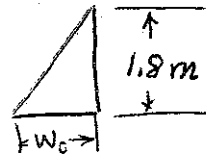
By writing $\sum F_y = 0$ and $\sum M_z = 0$ we can find the magnitude and point of action of the resultant force between upright and plank. The distribution cannot be determined by simple statics. However, since we

need to know the distribution to calculate M and V in the uprights, we will assume that it is the same as the water pressure distribution.

Thus if w = load per foot of upright, then

$$w = Cp \quad \text{where } p = \text{water pressure. } p = \gamma x$$

$$w = w_0 (x/1.8)$$



TOTAL FORCE ON 2.4×1.8 SECTION

$$\gamma_w 1.8 \times \frac{1}{2} \times 2.4 \times 1.8 = w_0 1.8 \times \frac{1}{2}$$

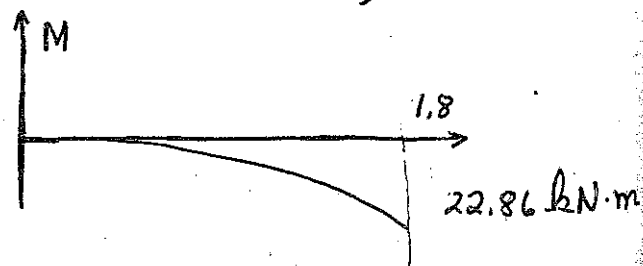
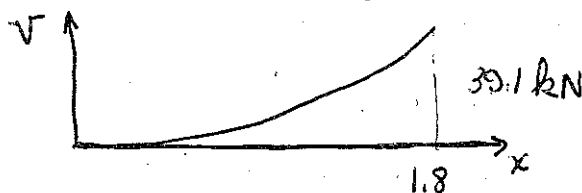
$$\gamma_w = 9.8 \text{ kN/m}^3$$

$$w_0 = \gamma_w 1.8 \times 2.4 = 42.34 \text{ kN/m}$$

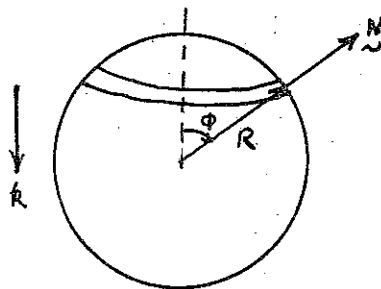
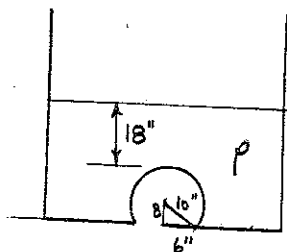
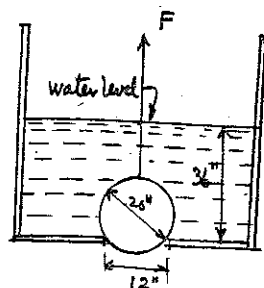
$$\text{Thus } q(x) = -w_0 x/1.8 = -23.52x$$

$$V(x) = -\int q(x) dx = 11.76x^2 \text{ (kN)}$$

$$M(x) = -\int V dx = -3.92x^3 \text{ (kN}\cdot\text{m)}$$



(335)



$$dF = -p n dA = -p n R^2 \sin \phi d\theta d\phi$$

$$F = -2\pi \int p n R^2 \sin \phi d\phi$$

$$F \cdot k = 2\pi R^2 \int p \sin \phi \cos \phi d\phi = F_k(t) \quad \text{where } n \cdot k = -\cos \phi$$

$$\phi = \left\{ 1.5 + \frac{5}{6}(1 - \cos \phi) \right\} \rho = \left\{ a + b(1 - \cos \phi) \right\} \rho$$

$$F_R = 2\pi R^2 \int_0^{\phi_0} \left\{ a + b(1 - \cos \phi) \right\} \rho \sin \phi \cos \phi d\phi$$

$$= \rho 2\pi R^2 \left[\frac{(a+b)}{2} \int_0^{\phi_0} \sin 2\phi d\phi - b \int_0^{\phi_0} \sin \phi \cos^2 \phi d\phi \right]$$

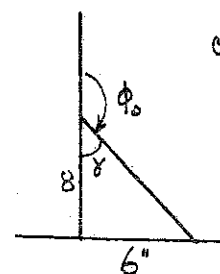
$$= \rho 2\pi R^2 \left[\frac{a+b}{2} \left\{ \frac{1 - \cos 2\phi_0}{2} \right\} - b \left\{ \frac{1 - \cos^3 \phi_0}{3} \right\} \right]$$

$$1 - \cos 2\phi_0 = 2 \sin^2 \phi_0$$

$$F_R = \rho 2\pi R^2 \left[\frac{a+b}{2} \sin^2 \phi_0 - b \left\{ \frac{1 - \cos^3 \phi_0}{3} \right\} \right] \quad \text{putting the values of } a, b, \sin^2 \phi_0, \cos^3 \phi_0 \text{ in,}$$

$$\text{we have } F_R = 0 !$$

For equilibrium $F = \text{weight of sphere} =$

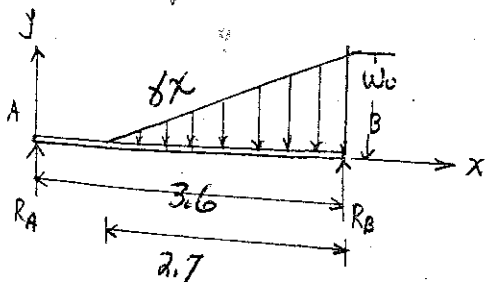


$$\begin{aligned} \cos \phi_0 &= \cos(\pi - \gamma) \\ &= -\cos \gamma \\ &= -\frac{4}{5} \end{aligned}$$

$$\begin{aligned} 1 - \cos^3 \phi_0 &= 1 + \frac{64}{125} \\ \sin^2 \phi_0 &= \frac{9}{25} \end{aligned}$$

-36) Assuming the fluid pressure varies linearly with depth.

Free-body diagram



$$p_B = \gamma \times 2.7$$

$$\text{FORCE} = p_B \times 2.7 \times 2.4 \times 0.5$$

$$\sum M_B = 0, \uparrow$$

$$F \times 0.9 = 3.6 R_A$$

$$R_A = 21.4 \text{ kN}$$

$$\omega = 1750(2\pi) \text{ rad/s}$$

3-37 Power Balance :

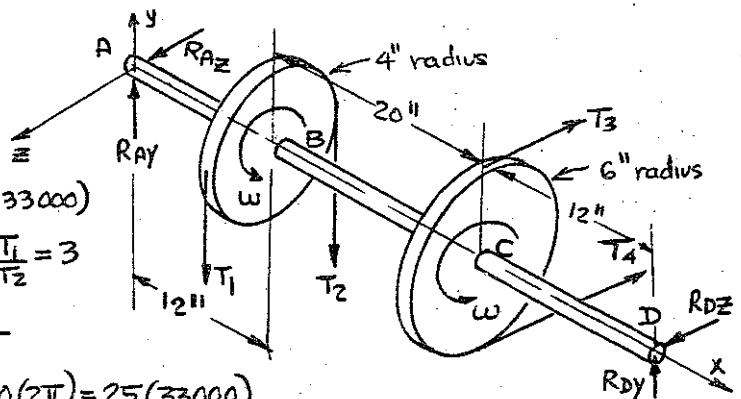
Pulley B

$$(T_1 - T_2) \left(\frac{4}{12} \right) 1750(2\pi) = 25(33000)$$

$$\therefore T_1 - T_2 = 226 \text{ lb and } \frac{T_1}{T_2} = 3$$

$$\therefore \underline{T_1 = 339 \text{ lb}}, \underline{T_2 = 113 \text{ lb}}$$

$$\underline{\text{Pulley C}} : (T_3 - T_4) \left(\frac{6}{12} \right) 1750(2\pi) = 25(33000)$$



$$T_3 - T_4 = 151 \text{ lb and } \frac{T_3}{T_4} = 3 \therefore \underline{T_3 = 226.5 \text{ lb}}, \underline{T_4 = 75.5 \text{ lb}}$$

Each pair of forces \$T_1, T_2\$ and \$T_3, T_4\$ is statically equivalent to a single force acting at the pulley center plus a moment

Pulley A : force = \$T_1 + T_2 = 452 \text{ lb}\$ (-y direction)

$$\text{Moment} = (T_1 - T_2) \left(\frac{4}{12} \right) = 75 \text{ lb-ft} = \underline{900 \text{ lb-in}} (+x \text{ direction})$$

Pulley B force = \$T_3 + T_4 = 302 \text{ lb}\$ (+z direction)

$$\text{Moment} = (T_3 - T_4) \left(\frac{6}{12} \right) = 75 \text{ lb-ft} = \underline{900 \text{ lb-in}} (-x \text{ direction})$$

Thus the forces and moments acting on the shaft are:

OVERALL EQUILIBRIUM :

$$\underline{\Sigma (M_z)_A = 0}$$

$$R_{Dy}(44) - 452(12) = 0$$

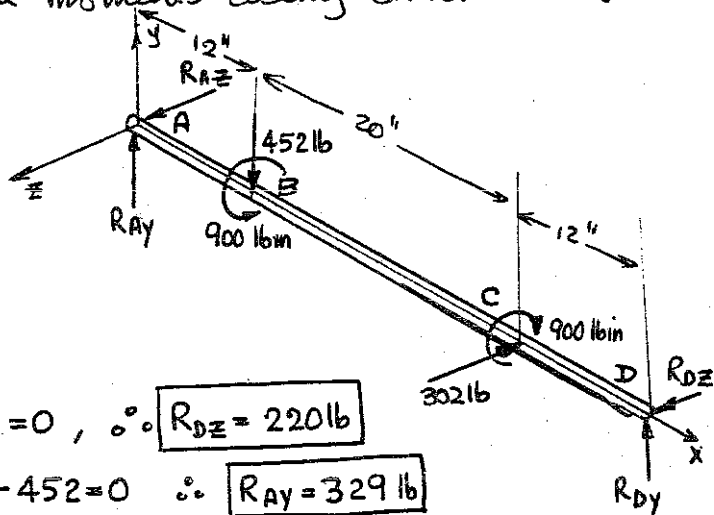
$$\therefore \underline{R_{Dy} = 123 \text{ lb}}$$

$$\underline{\Sigma (M_y)_A = 0}$$

$$-R_{Dz}(44) + 302(32) = 0, \therefore \underline{R_{Dz} = 220 \text{ lb}}$$

$$\underline{\Sigma F_y = 0} : R_{Dy} + R_{Ay} - 452 = 0 \therefore \underline{R_{Ay} = 329 \text{ lb}}$$

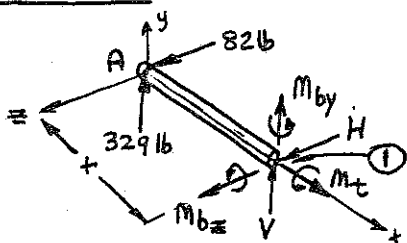
$$\underline{\Sigma F_z = 0} : R_{Dz} + R_{Az} - 302 = 0 \therefore \underline{R_{Az} = 82 \text{ lb}}$$



At each point in the shaft there will be a vertical (y) shear force \$V\$, a horizontal (z) shear force \$H\$, bending moments \$M_{by}\$ and \$M_{bz}\$ and a twisting moment \$M_t\$.

3-37 FREE BODY DIAGRAMS:

(cont'd) 1) $0 \leq x \leq 12''$



$$\sum F_y = 0 : V = -329 \text{ lb.}$$

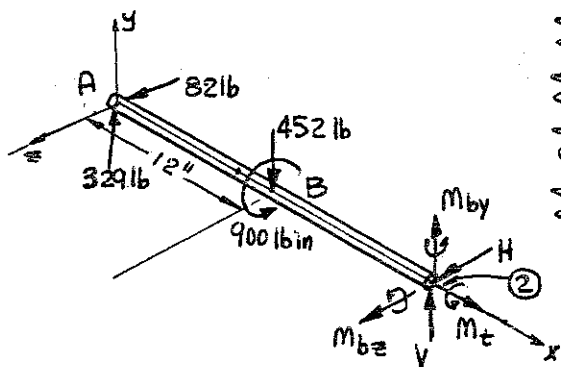
$$\sum F_x = 0 : H = -82 \text{ lb}$$

$$\sum (M_x)_1 = 0 : M_t = 0$$

$$\sum (M_y)_1 = 0 : M_{by} = -82x$$

$$\sum (M_z)_1 = 0 : M_{bz} = 329x$$

2) $12'' \leq x \leq 32''$



$$\sum F_y = 0 : V = 452 - 329 = 123 \text{ lb.}$$

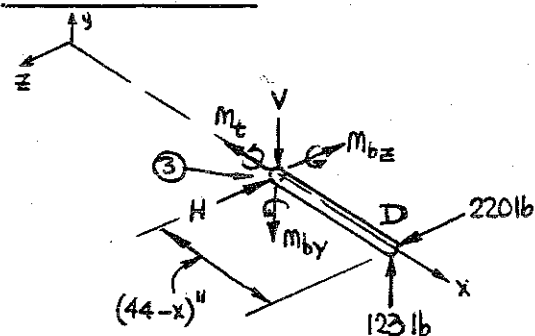
$$\sum F_x = 0 : H = -82 \text{ lb.}$$

$$\sum (M_x)_2 = 0 : M_t = -900 \text{ lb in.}$$

$$\sum (M_y)_2 = 0 : M_{by} = -82x \text{ lb in.}$$

$$\sum (M_z)_2 = 0 : M_{bz} = +329x - 452(x-12) \\ = (5424 - 123x) \text{ lb in}$$

3) $32'' \leq x \leq 44''$



$$\sum F_y = 0 : V = 123 \text{ lb.}$$

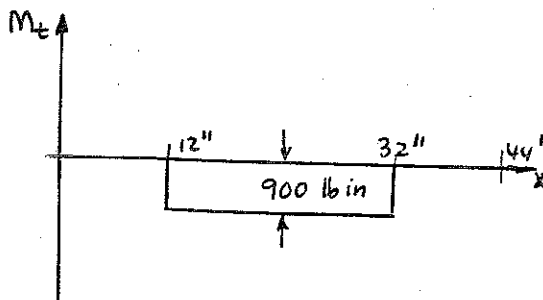
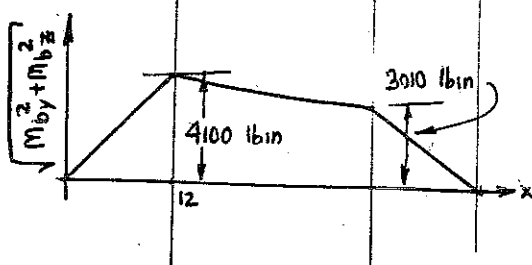
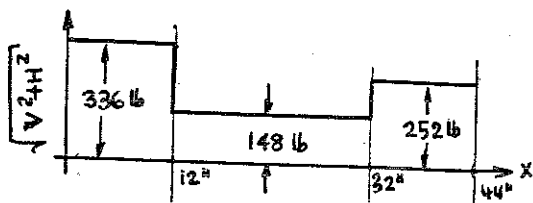
$$\sum F_x = 0 : H = 220 \text{ lb.}$$

$$\sum (M_x)_3 = 0 : M_t = 0$$

$$\sum (M_y)_3 = 0 : M_{by} = -220(44-x) \\ = (220x - 9680)$$

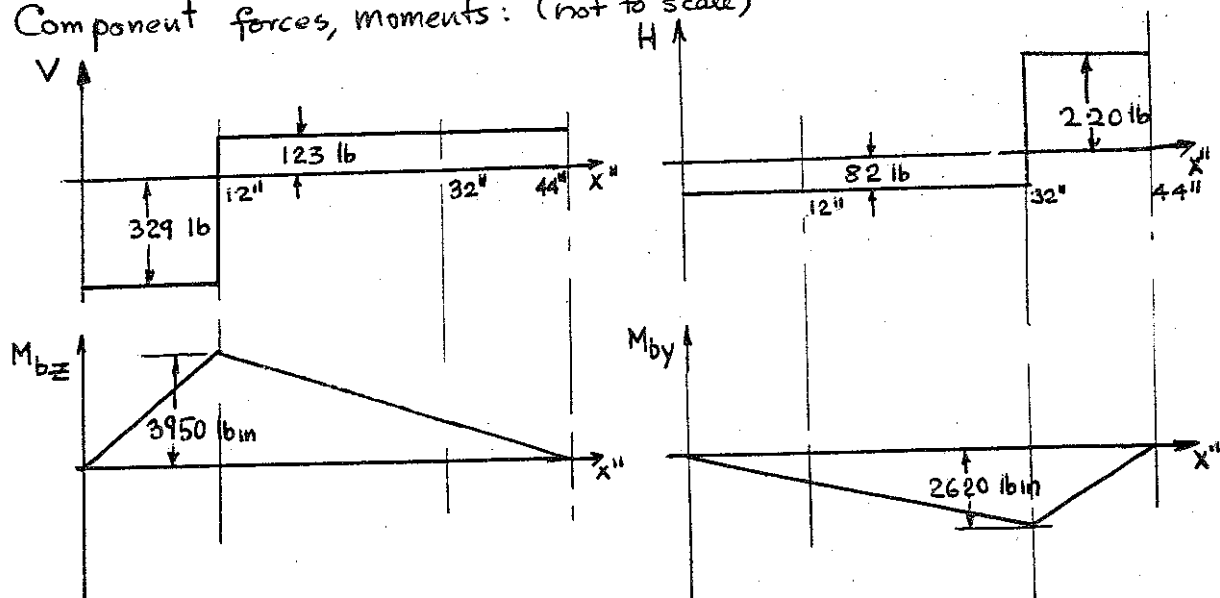
$$\sum (M_z)_3 = 0 : M_{bz} = 123(44-x) \\ = (5424 - 123x)$$

Plotted below are resultant shear force ($= \sqrt{V^2 + H^2}$), bending moment ($= \sqrt{M_{by}^2 + M_{bz}^2}$) and twisting moment (M_t). (Not to scale)

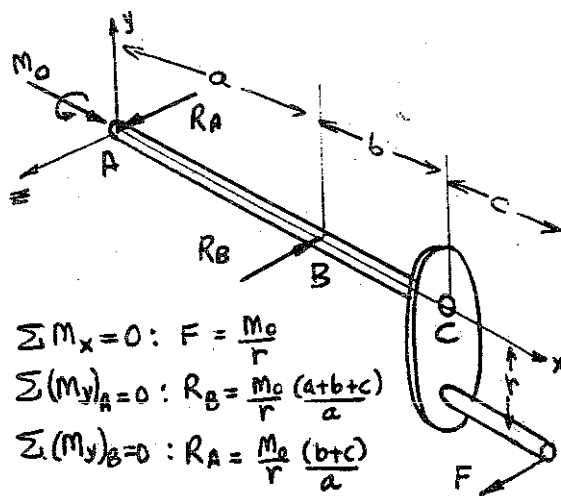


3-37 (cont'd)

Component forces, moments: (not to scale)



3.38

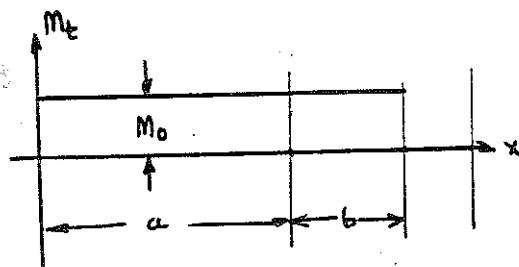


$$\sum M_x = 0: F = \frac{M_0}{r}$$

$$\sum (M_y)_A = 0: R_B = \frac{M_0}{r} \left(\frac{a+b+c}{a} \right)$$

$$\sum (M_y)_B = 0: R_A = \frac{M_0}{r} \left(\frac{b+c}{a} \right)$$

$\sum F_z = 0$: satisfied by above
 R_A, R_B, F

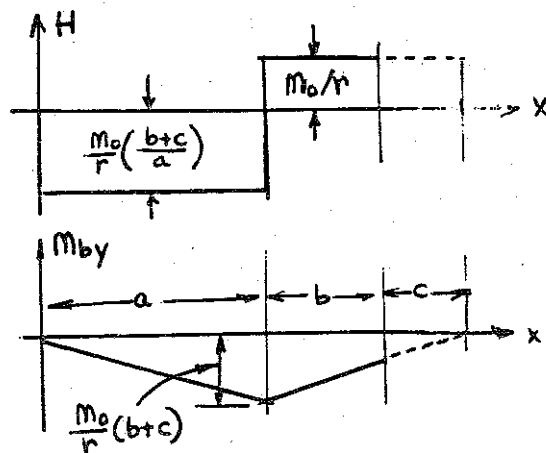


The only shear force is in the z direction $= H$

The twisting moment M_t is constant and equal to M_0

The bending moment M_{bz} is zero.

The bending moment M_{by} has its max. value at B, where the shear force passes through zero



3-39

Radius of smaller

$$r_{\text{gear}} = \frac{1}{3}(62.5) =$$

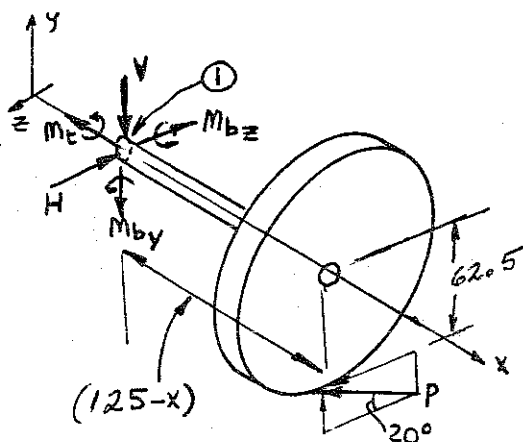
Small gear :

$$\sum (M_x) = 0 \text{ about gear center.}$$

$$(130 \text{ N}\cdot\text{m}) - P \cos 20^\circ (20.8 \times 10^{-3})$$

$$\therefore P = 5.87 \text{ kN}$$

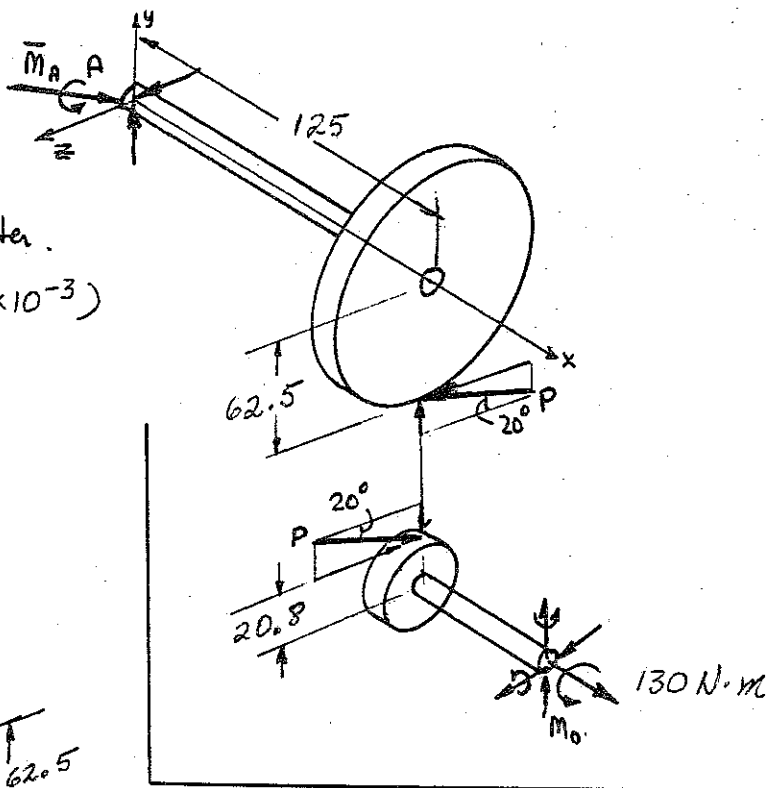
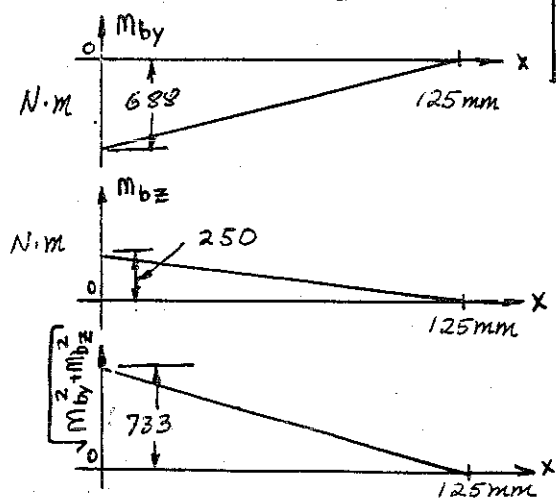
Free Body :



$$V = \text{constant} = 2.0 \text{ kN}$$

$$H = \text{constant} = 5.51 \text{ kN}$$

$$\sqrt{V^2 + H^2} = 5.86 \text{ kN}$$



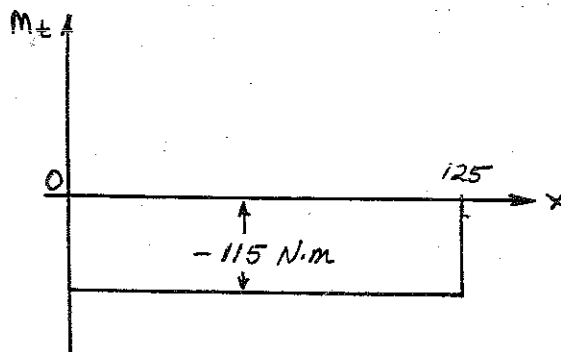
$$\sum F_y = 0 : V = P \sin 20^\circ = 2.0 \text{ kN}$$

$$\sum F_z = 0 : H = P \cos 20^\circ = 5.51 \text{ kN}$$

$$\sum (M_x)_1 = 0 : M_t = -(20.8) P \cos 20^\circ = -115 \text{ N}\cdot\text{m}$$

$$\sum (M_y)_1 = 0 : M_{by} = -(125-x) P \cos 20^\circ = -5.51(125-x)$$

$$\sum (M_z)_1 = 0 : M_{bz} = 2.0(125-x)$$



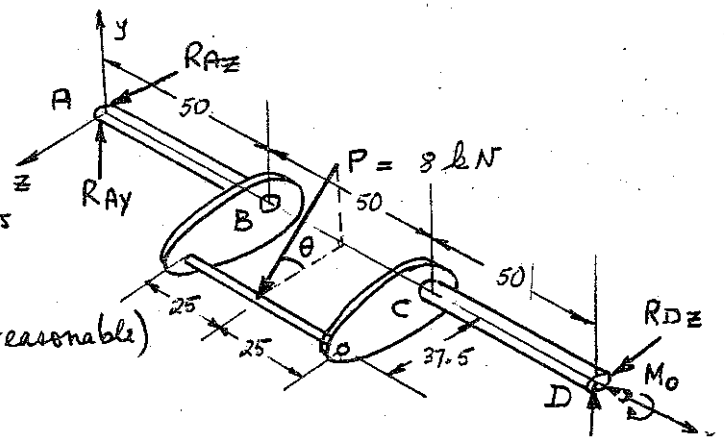
3.40

Crank radius = $\frac{1}{2}$ (stroke)
= 1.5 mm

Assume that bearings do not transmit moments to shaft.

Assume $\theta = 70^\circ$ (seems reasonable)

Overall equilibrium:



$$\sum M_x = 0: M_0 = (P \sin \theta) 37.5 = 8 (0.94) 37.5 = 282 \text{ N}\cdot\text{m}$$

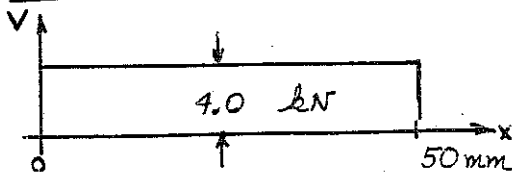
$$\sum (M_y)_A = 0: -R_{Dz} (150) - P \cos \theta (75) = 0 \quad R_{Dz} = -1.37 \text{ kN}$$

$$\sum (M_z)_A = 0 \quad R_{Dy} (150) - P \sin \theta (75) = 0 \quad R_{Dy} = 3.76 \text{ kN}$$

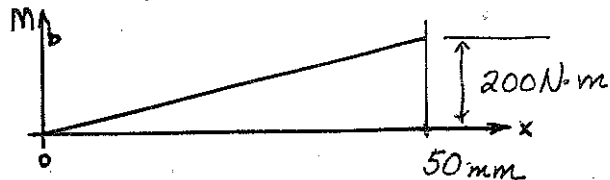
Symmetry: $R_{Ax} = R_{Dz}$; $R_{Ay} = R_{Dy}$

It is only necessary to sketch the shear force and bending moment diagrams for AB. That for CD is identical, except that the sign of V is reversed.

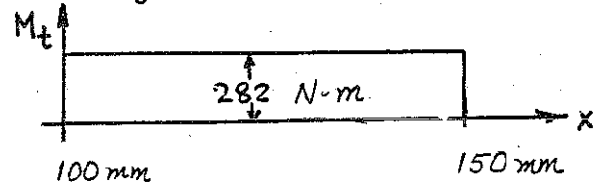
Shear force, AB, CD



Bending Moment, AB, CD



Twisting moment, CD only



Twisting moment in AB = 0.

3.41 For a close-coiled spring we may assume that the wire axis is perpendicular to the spring axis. We will form a free body by making a cut normal to the wire axis.

a) Refer to Fig 1:

Force equilibrium requires that there be a vertical force P acting on the cross section.

Moment equilibrium requires that there be a moment on the cross section that balances the moment of the applied force P . This moment can be resolved into two components: one parallel to the wire axis (twisting moment) and the other perpendicular to it (bending moment). As can be seen from the figure: $M_t = PR$, $M_b = 0$

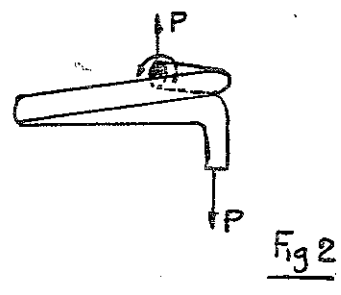
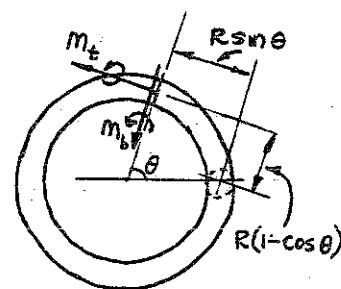
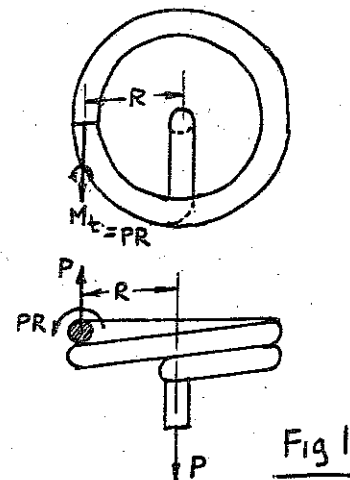
independent of the location of the cross section.

b) Figure 2:

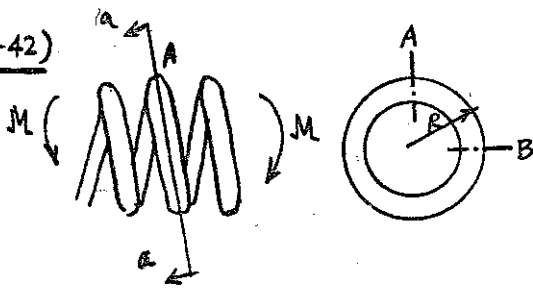
Here we obtain $M_t = PR(1 - \cos\theta)$

$M_b = PR \sin\theta$

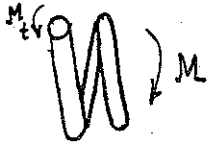
} see figure



(3-42)



Free-body dia. with a cut across A.

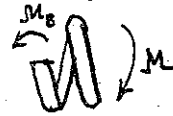


For moment equilibrium,

$$M_t = M \quad (\text{twisting moment})$$

For force equilibrium, since there is no external forces acting, All forces vanish.

Free-body dia with cut across B



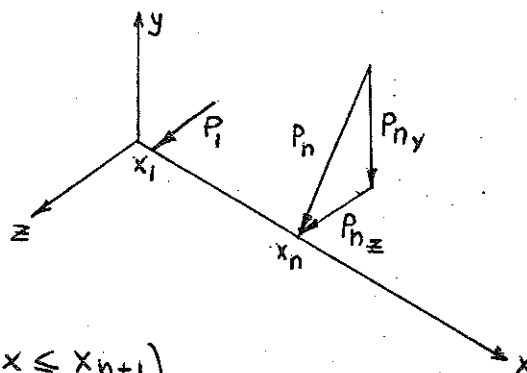
For moment equilibrium,

$$M_b = M \quad (\text{bending moment})$$

For force equilibrium, since there is no external forces acting, there is no force acting at the cut B.

3.43

Let there be N forces, the n^{th} force acting at x_n and having components P_{ny}, P_{nz} as shown. We will develop expressions for M_{bz} and M_{by} valid over the interval $(x_n \leq x \leq x_{n+1})$.



$$M_{by} = \sum (\text{moments of all } P_{nz} \text{ from } r=1 \text{ to } r=n) \\ = \sum_{r=1}^n P_{rz} (x - x_r) = x \sum_{r=1}^n P_{rz} - \sum_{r=1}^n P_{rz} x_r = Ax + B$$

$$M_{bz} = \sum_{r=1}^n P_{ry} (x - x_r) = Cx + D$$

where A, B, C, D are constant over $(x_n \leq x \leq x_{n+1})$

Then M_{by} and M_{bz} are represented by straight line segments and the maximum must occur at either x_n or x_{n+1} , i.e. at a loading point.

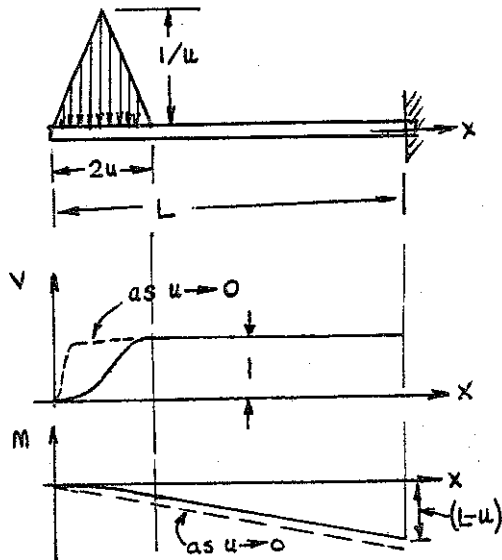
$$M_b = \sqrt{M_{by}^2 + M_{bz}^2} = \sqrt{(Ax+B)^2 + (Cx+D)^2} \\ = \sqrt{(A^2+C^2)x^2 + 2(AB+CD)x + (B^2+D^2)}, \text{ and } \underline{\underline{M_b > 0}} \\ = \sqrt{ax^2 + bx + c}$$

$$\frac{d^2 M_b}{dx^2} = \frac{(4ac - b^2)}{4(ax^2 + bx + c)^{3/2}} = \frac{4(A^2+C^2)(B^2+D^2) - 4(AB+CD)^2}{4[(A^2+C^2)x^2 + 2(AB+CD)x + B^2+D^2]^{3/2}} \quad (1)$$

The sign of $\frac{d^2 M_b}{dx^2}$ is the sign of the numerator of (1)

$$4(A^2+C^2)(B^2+D^2) - 4(AB+CD)^2 = 4(AD-BC)^2 \text{ which is always positive.} \\ \therefore \frac{d^2 M_b}{dx^2} \geq 0.$$

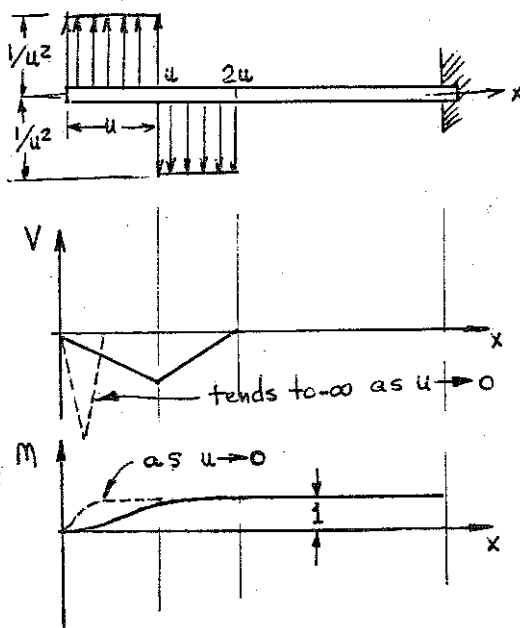
3.44



As can be seen from the V diagram, as $u \rightarrow 0$, $q_{-1}(x, u)$ approaches the step function $\langle x \rangle^0$. This corresponds to a loading $q(x) = -\frac{d}{dx} V(x) = -\langle x \rangle_1$, which is a concentrated downward load at $x=0$.

Similarly $M(x) \Rightarrow -\langle x \rangle^1$ and $q(x) = \frac{d^2}{dx^2} M(x) = -\langle x \rangle_1$, as before.

3.45



As $u \rightarrow 0$ the shear force goes to $-\infty$ over an infinitesimal length, and is zero for $x > 0$.

The bending moment approaches the function $\langle x \rangle^0$ as $u \rightarrow 0$, and the loading approaches a concentrated couple, of unit moment, at the origin.

$$q(x) = \frac{d^2}{dx^2} M(x) = \frac{d^2}{dx^2} \langle x \rangle^0 = \langle x \rangle_2$$

$$q_{-2}(x, u) = \begin{cases} 0 & \text{for } -\infty < x < 0 \\ 1/u^2 & \text{for } 0 < x < u \\ -1/u^2 & \text{for } u < x < 2u \\ 0 & \text{for } 2u < x < \infty \end{cases}$$

$$q_{-1}(x, u) = \begin{cases} 0 & \text{for } -\infty < x < 0 \\ x/u^2 & \text{for } 0 < x < u \\ \frac{2}{u} - \frac{x}{u^2} & \text{for } u < x < 2u \\ 0 & \text{for } 2u < x < \infty \end{cases}$$

For $-\infty < x < 0$: $\int_{-\infty}^x q_{-2}(x, u) dx = 0.$

For $0 < x < u$: $\int_{-\infty}^x q_{-2}(x, u) dx = \int_{-\infty}^0 q_{-2} dx + \int_0^x q_{-2} dx = 0 + \int_0^x \frac{1}{u^2} dx = \left(\frac{x}{u^2}\right)$

For $u < x < 2u$: $\int_{-\infty}^x q_{-2} dx = \int_{-\infty}^0 q_{-2} dx + \int_0^u q_{-2} dx + \int_u^x q_{-2} dx$
 $= 0 + \frac{u}{u^2} + \int_u^x \left(-\frac{1}{u^2}\right) dx = \frac{1}{u} - \frac{x}{u^2} + \frac{1}{u}$
 $= \left(\frac{2}{u} - \frac{x}{u^2}\right)$

For $2u < x < \infty$ $\int_{-\infty}^x q_{-2} dx = \left(\frac{2}{u} - \frac{2u}{u^2}\right) + 0 = 0.$

Thus

$$\int_{-\infty}^x q_{-2}(x, u) dx = \begin{cases} 0 & \text{for } -\infty < x < 0 \\ \frac{x}{u^2} & \text{for } 0 < x < u \\ \frac{2}{u} - \frac{x}{u^2} & \text{for } u < x < 2u \\ 0 & \text{for } 2u < x < \infty \end{cases}$$

This is precisely the definition of $q_{-1}(x, u)$

$\therefore \int_{-\infty}^x q_{-2}(x, u) dx = q_{-1}(x, u)$ Also $\lim_{u \rightarrow 0} q_{-2}(x, u) = \langle x \rangle_{-2}$
and $\lim_{u \rightarrow 0} q_{-1}(x, u) = \langle x \rangle_{-1}$

$\lim_{u \rightarrow 0} \int_{-\infty}^x q_{-2}(x, u) dx = \lim_{u \rightarrow 0} q_{-1}(x, u) = \langle x \rangle_{-1}$

$\lim_{u \rightarrow 0} \int_{-\infty}^x q_{-2}(x, u) dx = \int_{-\infty}^x \left\{ \lim_{u \rightarrow 0} q_{-2}(x, u) \right\} dx = \int_{-\infty}^x \langle x \rangle_{-2} dx.$

$\therefore \int_{-\infty}^x \langle x \rangle_{-2} dx = \langle x \rangle_{-1}$ or, in general, $\int_{-\infty}^x \langle x-a \rangle_{-2} dx = \langle x-a \rangle_{-1}$